Equilibria and Recurrent Dynamics of Normalized Transformer Blocks

Exact stationary equations, symmetry boundaries, and a reproducible numerical audit

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Abstract

We study the dynamics obtained by repeatedly applying a normalized residual self-attention layer followed by a normalized residual quadratic MLP. In the continuous-depth equation, a configuration is stationary exactly when the total attention–MLP output at each token is radial. Without the MLP, this condition admits an equivalent spectral–Gram formulation under the tied symmetry $Q^{\mathsf{T}}K = V = V^{\mathsf{T}}$; it covers all eigenvalue multiplicities, sign patterns, mixed eigenspace configurations, and zero modes. The same symmetry yields a strict Lyapunov function and excludes nonconstant recurrent trajectories. Untying score and value transport removes this obstruction. For one explicit three-token system, long integration, tangent Lyapunov spectra, and perturbation tests give numerical evidence for an attracting internal periodic orbit under attention alone.

For the serial block, the continuous-depth limit is the projected sum of the attention and MLP fields; layer ordering first appears at second order in the layer strength. A census of 28,672 finite-layer models and 2,752,512 trajectories finds stable period-three and period-four orbits in each of four symmetry classes. A separate weak-layer audit samples 214,528 models and 2,844,160 trajectories. Among 1,908 selected attractors, 1,699 approach equilibria and 209 remain mobile on the main observation window. Direct integration and tangent-spectrum tests isolate stable internal cycles, two three-token chaotic-attractor candidates, and one eight-token candidate with three positive Lyapunov exponents. The reciprocal-attention, potential-MLP class has no recurrent example in 286 tested cases; this is an empirical statement, whereas recurrence is excluded analytically only in the identified Lyapunov regimes.

We also examine multihead cancellation, high-dimensional score scaling, the many-token transport equation, alternative attention kernels, trained MLP control, one-step spectral optimization, and slowly varying random weights. In an exactly solvable neural-quadratic-form control, all 280 sign and unordered-multiplicity classes through dimension six satisfy the predicted initial growth rates; three initialization laws retain the predicted logarithmic ignition slopes. The tested BBP threshold occurs in the finite-width Gaussian covariance ensemble; the row-Stiefel control has no Wishart bulk. These extensions separate the state dynamics at inference from the weight dynamics during training.

A 30,656-trajectory settling-time atlas tests a recent fractal interpretation of recurrent reasoning. Co-moving zooms near a synchronized saddle–node give fitted uncertainty slopes from 1.03 to 0.66. At the deepest zoom, the set excluded by the combined speed-and-root-distance audit remains one strip whose affine-edge residual is below 0.30 pixel RMS. All full-interval tangent exponents for the three audited trajectories are negative. The local fold normal form gives an uncertainty exponent 1/2 for a smooth singular set, while its hyperbolic continuation approaches exponent one only after fixed-window resolution refinement. At the tested scales, the observed map quantitatively matches critical slowing and does not establish a fractal boundary.

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1 Introduction

Repeated self-attention defines a nonlinear dynamical system on token configurations. Existing analyses establish clustering, consensus, and spectral selection under structural assumptions on the score and value matrices [22, 13, 6, 29]. Those assumptions are consequential: they determine whether the flow admits a monotone energy, which stationary configurations are available, and whether recurrent motion can occur. A Transformer block also contains a serial MLP, while practical models use several separately normalized heads. The resulting vector field need not retain the reciprocal structure of a single symmetric head.

This paper studies one elementary model in which these distinctions can be made exactly. Tokens lie on a product of spheres. A residual attention update is renormalized, followed by a residual quadratic MLP update and a second normalization. The model retains the score matrix $Q^{\mathsf{T}}K$, value transport, row-wise softmax, serial ordering, and tokenwise normalization while remaining small enough for exact stationary equations and complete tangent calculations.

The analytical results have three scopes. For arbitrary score and value matrices, the attention-only equilibrium equation is necessary and sufficient. Under $Q^{\mathsf{T}}K = V = V^{\mathsf{T}}$ it reduces to a positive-semidefinite Gram system indexed by the eigenspaces of V, including repeated and zero eigenvalues. In the same tied case a strict Lyapunov identity excludes cycles. For the serial block, the finite-layer fixed-point and periodic-orbit equations remain exact, and the continuous-depth limit is obtained with an explicit first-order error statement.

The numerical work addresses behavior that these identities permit but do not classify. A finite-layer census is followed by continuation toward weak layers, direct ODE integration, complete Lyapunov spectra, perturbation tests, and basin surveys. The resulting examples give numerical evidence for stable internal cycles and for long-lived trajectories with the tangent-spectrum signatures of chaos. The experiments report existence and conditional frequencies on specified sampling schemes; they do not imply prevalence under an unspecified random-weight law.

Later sections treat changes of scale and architecture. Several heads can cancel in tangent directions and create equilibria absent from every head in isolation. Near-orthogonality in high dimension does not by itself decouple tokens, and the many-token limit is a nonlinear transport equation. Training and slowly varying weights act on a second time scale. We compare our fixed-weight dynamics with training-induced escape from clusters, neural quadratic-form mode ignition, and recent observations of transient chaos in recurrent reasoners. The final comparison distinguishes persistent attractors, smooth multistability, and fractal settling-time fields.

Several nearby results delimit these questions. Tomihari and Karakida analyze normalized recurrent multihead self-attention through its Jacobian and finite-time Lyapunov exponents; their experiments already report mildly positive exponents [44]. Burger et al. construct mixed stationary measures for tied spherical attention [13], and Gao, Yang, and Chen derive high-dimensional clustered manifolds in a different joint scaling [20]. Polyanskiy, Rigollet, and Yao prove almost-everywhere synchronization for the homogeneous reciprocal circle model at all $\beta \geq 0$ [38]. These works preclude claims that mixed equilibria, Jacobian diagnostics, or oscillatory attention had not appeared previously. The present analytical contribution is the complete finite-token spectral–Gram reduction under tied symmetry, the finite serial fixed-point and orbit equations, the order-h^2 ordering calculation, and exact recurrent constructions outside the Lyapunov regimes. The numerical contribution is the census and its weak-layer audit. The more complicated attractors are reported with numerical evidence and explicit diagnostics, without a theorem of existence. A separate multiscale atlas gives a negative result: a visually sharp settling-time field near a saddle–node is smooth at the tested resolution, and its low fitted uncertainty slopes are reproduced by nonfractal local normal forms.

The convergence literature mainly treats reciprocal or isotropic sectors. The original continuous

attention model exhibits cluster formation [22]; in the corresponding spherical gradient regime, multicluster states can persist for exponentially long times before eventual collapse [21]. Bruno, Pasqualotto, and Agazzi identify metastable mean-field modes near the uniform law through spherical harmonics [12], while Chen, Lin, Polyanskiy, and Rigollet prove exponential convergence to a Dirac mass for regular initial laws under their synchronizing parameter assumptions [15]. Karagodin, Ge, Polyanskiy, and Rigollet compare normalization schemes by the speed regulation they induce in this clustering flow [27]. Our stable mixed equilibrium uses anisotropic tied transport and does not contradict the homogeneous consensus results; the recurrent examples leave reciprocity altogether.

Other limits evolve the parameters as well as the tokens. Barboni, de Hoop, Furuya, and Peyré formulate infinite-depth and infinite-head training as a conditional Wasserstein gradient flow and prove local or small-loss convergence under an injective attention NTK condition [9]. Liu and Herty couple a token transport equation to a Wasserstein parameter flow and obtain well-posedness, an entropy-regularized shallow-model result, and local deep-model convergence under NTK nondegeneracy [31]. Those are training-time theorems. The equilibrium and recurrent analysis below freezes S, V, M and evolves the token state through repeated layers.

Feed-forward and positional extensions change the stationary set in ways that cannot be represented by a different spectrum of V alone. Álvarez-López, Geshkovski, and Ruiz-Balet prove localization and generic atomicity results, under their activation and parameter assumptions, for stationary measures in a mean-field attention model with a perceptron potential [7]. Imaizumi, Koyama, Isobe, and Hayashi show that a fixed auxiliary variable, including a positional or prefix variable, can retain a noncollapsed marginal law even when the conditional maximizing law is a Dirac mass [25]. For query/key rotary position embeddings, Ye derives position-locked twisted and antipodal equilibria and their tangent spectra [47]. Our token-only equations use the same S for every pair and do not cover those position-dependent score kernels.

2 Model

2.1 Tokens and attention

There are n tokens. Each token is a unit vector $x_i \in \mathbb{R}^d$. The two matrices actually visible to one attention head are

$$S = Q^\mathsf{T} K, \qquad V.$$

S decides who attends to whom, and V decides what is transported. Frozen dynamics never sees Q and K separately: two factorizations with the same product S produce exactly the same trajectories.

For sharpness $\beta \geq 0$, set

$$a_{ij}(X) = \frac{\exp(\beta x_i^\mathsf{T} S x_j)}{\sum_{k=1}^n \exp(\beta x_i^\mathsf{T} S x_k)}, \tag{1}$$

$$A_i(X) = \sum_{j=1}^n a_{ij}(X) V x_j. \tag{2}$$

Even at $\beta = 0$, attention has not been removed: every weight equals $1/n$, and V still transports the token mean.

2.2 The quadratic MLP

The smallest MLP that already produces multiple wells and circulation is

$$M(x) = b + Lx + \sum_{r=1}^{m} o_r \, (u_r^\mathsf{T} x + c_r)^2. \tag{3}$$

Its width is m. Vector u_r determines what hidden neuron r measures, and o_r determines the direction in which its response pushes.

We call this MLP *potential* when

$$L = L^\mathsf{T}, \qquad o_r = \gamma_r u_r. \tag{4}$$

Then there is a scalar function

$$\phi(x) = b^\mathsf{T} x + \tfrac{1}{2} x^\mathsf{T} L x + \sum_r \frac{\gamma_r}{3} (u_r^\mathsf{T} x + c_r)^3 \quad \text{such that} \quad M = \nabla\phi. \tag{5}$$

This constraint does not make the MLP weak: it can create several wells. It only rules out a circulating push that is not the slope of one common surface.

Proposition 2.1 (Exact planar capacity of a quadratic MLP)**.** *On $\mathbb{S}^1$, write $x(\theta) = (\cos\theta, \sin\theta)$ and $t(\theta) = (-\sin\theta, \cos\theta)$. The angular field $m(\theta) = t(\theta)^\mathsf{T} M(x(\theta))$ of a general quadratic MLP is a real trigonometric polynomial of degree at most three. Conversely, with sufficient width, every*

$$c_0 + \sum_{q=1}^{3} (a_q \cos(q\theta) + b_q \sin(q\theta))$$

is realized by such an MLP. If $M = \nabla\phi$ is potential, the realizable fields are exactly those in this class with $c_0 = 0$.

Proof. Each component of $M(x)$ is a polynomial of degree at most two. Multiplication by the components of $t(\theta)$ and the standard product-to-sum identities give harmonics of degree at most three. Conversely, vector-valued quadratic polynomials span these seven harmonics, and rank-one squares $(u^\mathsf{T} x + c)^2$ span scalar quadratics. In the potential case, $m(\theta) = d[\phi(x(\theta))]/d\theta$, so its mean over the circle is zero. Every zero-mean trigonometric polynomial of degree at most three is the derivative of the restriction of a cubic polynomial; cubes of affine linear forms span polynomials of total degree at most three after homogenization. □

Unless the angular field vanishes identically, one isolated token therefore has at most six equilibria and at most three asymptotically stable ones. Potentiality removes the constant circulation mode but retains the three-well construction. Token coupling can exceed this single-token count because the exponential attention field depends on the full configuration.

2.3 Serial layer

Let $\mathcal{N}(z) = z/\|z\|$. A layer of strength $h > 0$ is

$$y_i = \mathcal{N}(x_i + hA_i(X)), \qquad F_i(X) = \mathcal{N}(y_i + hM(y_i)). \tag{6}$$

The MLP is genuinely *after* attention, as in the standard Transformer block order [45]. By contrast, the control model of Isobe, Inoue, and Imaizumi adds an FFN field to the attention field in a continuous equation [26]. This architectural difference matters at finite step size; the next section states what becomes of it as h tends to zero.

3 Attention-only stationary configurations

Set $M = 0$ and consider the continuous attention equation

$$\dot{x}_i = (I - x_i x_i^\mathsf{T}) A_i(X), \qquad i = 1, \dots, n. \tag{7}$$

Its stationary configurations can be described without symmetry, definiteness, or a genericity assumption.

Theorem 3.1 (Equilibrium equation for arbitrary score and value matrices)**.** *A configuration $X \in (\mathbb{S}^{d-1})^n$ is stationary for (7) if and only if, for each token i, there is a scalar λ_i such that*

$$A_i(X) = \lambda_i x_i, \qquad \lambda_i = x_i^\mathsf{T} A_i(X). \tag{8}$$

Equivalently,

$$\sum_{j=1}^{n} \exp(\beta x_i^\mathsf{T} S x_j) \left[V x_j - (x_i^\mathsf{T} V x_j) x_i \right] = 0, \qquad i = 1, \dots, n. \tag{9}$$

Proof. The right-hand side of (7) vanishes exactly when $A_i(X)$ belongs to the line spanned by x_i. Its coefficient on that line is $x_i^\mathsf{T} A_i(X)$. Multiplication by the strictly positive softmax denominator gives (9). □

The output need not vanish. Spherical normalization removes every radial component, so a noncollinear configuration can be stationary with $A_i(X) \neq 0$ for every token. This distinction is required when comparing with Altafini's polygonal class, which is defined by zero received output.

If the configuration contains q distinct positions $w_1, \dots, w_q$ with multiplicities $r_1, \dots, r_q$, (9) is equivalent to the finite system

$$\sum_{b=1}^{q} r_b \exp(\beta w_a^\mathsf{T} S w_b) \left[V w_b - (w_a^\mathsf{T} V w_b) w_a \right] = 0, \qquad a = 1, \dots, q. \tag{10}$$

This is an exact grouping of identical terms. For fixed n, d, S, V, enumeration of the integer partitions $r_1 + \cdots + r_q = n$ followed by solution of (10) covers all clustered branches.

3.1 Tied symmetric attention

Assume in this subsection that

$$S = V = V^\mathsf{T}. \tag{11}$$

This is the structural relation used by Kuehn and Yoon [29]. It is stronger than symmetry of the value matrix: Altafini allows a general $Q^\mathsf{T} K$ while imposing hypotheses on V [6]. Burger, Kabri, Korolev, Roith, and Weigand analyze the corresponding spherical mean-field energy for tied choices $V = \pm S$ and construct mixed atomic stationary measures, including a non-eigenvector four-atom rectangle [13]. The mixed family below is therefore a finite-token compatibility calculation, not the first non-eigenvector stationary measure in the tied model.

Write the tokens as the rows of $X \in \mathbb{R}^{n \times d}$ and define

$$H = XVX^\mathsf{T}, \qquad R = \exp_\circ(\beta H), \qquad \Lambda_X = \mathrm{Diag}(\mathrm{diag}(RH)), \tag{12}$$

where $\exp_\circ$ denotes entrywise exponentiation.

Theorem 3.2 (Vector and spectral–Gram characterizations)**.** *Under (11), X is an equilibrium if and only if*

$$RXV = \Lambda_X X, \qquad \operatorname{diag}(XX^\mathsf{T}) = \mathbf{1}. \tag{13}$$

Let $\nu_1, \dots, \nu_s$ be the distinct eigenvalues of V, with multiplicities $m_1, \dots, m_s$. Write the component of X in the ν_α eigenspace as $C_\alpha \in \mathbb{R}^{n \times m_\alpha}$ and set $G_\alpha = C_\alpha C_\alpha^\mathsf{T}$. Modulo orthogonal changes of basis inside each eigenspace, the equilibria are exactly the families satisfying

$$G_\alpha \succeq 0, \qquad \operatorname{rank} G_\alpha \le m_\alpha, \qquad \sum_{\alpha=1}^{s} \operatorname{diag} G_\alpha = \mathbf{1}, \tag{14}$$

$$H = \sum_{\alpha=1}^{s} \nu_\alpha G_\alpha, \qquad R = \exp_\circ(\beta H), \qquad \Lambda_X = \operatorname{Diag}(\operatorname{diag}(RH)), \tag{15}$$

$$(\nu_\alpha R - \Lambda_X) G_\alpha = 0, \qquad \alpha = 1, \dots, s. \tag{16}$$

Proof. Removing the row denominators in (8) gives $RXV = \Lambda_X X$; its ith diagonal coefficient is $(\Lambda_X)_{ii} = \sum_j R_{ij} x_i^\mathsf{T} V x_j = (RH)_{ii}$. In an eigenbasis of V, projection onto the ν_α eigenspace gives $(\nu_\alpha R - \Lambda_X) C_\alpha = 0$. Multiplication by C_α^T yields (16); the other identities follow from the orthogonal eigenspace decomposition.

Conversely, factor each feasible G_α as $C_\alpha C_\alpha^\mathsf{T}$ using at most m_α columns. The image of G_α equals the image of C_α, so (16) implies $(\nu_\alpha R - \Lambda_X) C_\alpha = 0$. Placing the factors in mutually orthogonal eigenspaces reconstructs a unit-row matrix X satisfying (13). Any two factors $C_\alpha, \widetilde{C}_\alpha$ with $C_\alpha C_\alpha^\mathsf{T} = \widetilde{C}_\alpha \widetilde{C}_\alpha^\mathsf{T}$ are related by an orthogonal map on the ambient eigenspace: the isometry between their row spaces extends orthogonally to the full eigenspace. □

The tied symmetry also supplies the obstruction to recurrent motion claimed in the introduction. For $\beta > 0$, define

$$\mathcal{E}_\beta(X) = \frac{1}{2\beta} \sum_{i,j=1}^{n} \exp\left(\beta x_i^\mathsf{T} V x_j\right). \tag{17}$$

Since V and R are symmetric, differentiation along (7) gives the exact identity

$$\frac{d}{dt} \mathcal{E}_\beta(X(t)) = \sum_{i=1}^{n} Z_i(X(t)) \, \|\dot{x}_i(t)\|^2, \qquad Z_i = \sum_j \exp(\beta x_i^\mathsf{T} V x_j) > 0. \tag{18}$$

At $\beta = 0$, the corresponding finite energy is

$$\mathcal{E}_0(X) = \frac{1}{2n} \left(\sum_i x_i \right)^\mathsf{T} V \left(\sum_i x_i \right), \qquad \frac{d}{dt} \mathcal{E}_0 = \sum_i \|\dot{x}_i\|^2. \tag{19}$$

Thus every nonstationary trajectory strictly increases the relevant energy. In particular, the tied symmetric attention flow has no nonconstant periodic or recurrent orbit.

Proposition 3.3 (Absence of zero-output polygons for $V \succ 0$)**.** *Assume $S = V \succ 0$ and $\beta > 0$. No unit-token configuration has zero received output at every distinct token position.*

Proof. Group the distinct positions as $w_1, \dots, w_q$ with positive multiplicities $r_1, \dots, r_q$, and let W have rows w_a^T. The zero-output equations are

$$\mathcal{R} \operatorname{Diag}(r_1, \dots, r_q) W V = 0, \qquad \mathcal{R}_{ab} = \exp\left(\beta \langle V^{1/2} w_a, V^{1/2} w_b \rangle\right).$$

The exponential dot-product kernel is strictly positive definite on distinct points, so $\mathcal{R}$ is invertible. The multiplicity matrix and V are also invertible, which would force $W = 0$, contradicting its unit rows.

For completeness, put $z_a = V^{1/2} w_a$. For any coefficients c_a,

$$\sum_{a,b} c_a c_b \mathrm{e}^{\beta z_a^{\mathsf{T}} z_b} = \sum_{m=0}^{\infty} \frac{\beta^m}{m!} \left\| \sum_a c_a z_a^{\otimes m} \right\|^2 .$$

If this expression vanishes, every tensor sum on the right vanishes. Choose a vector v for which the distinct points z_a have distinct projections $v^{\mathsf{T}} z_a$. Contracting the first q tensor identities with $v^{\otimes m}$ gives a Vandermonde system, hence every $c_a = 0$. □

The restriction $\beta > 0$ is necessary. At $\beta = 0$, $\mathcal{R}$ has rank one and every zero-sum configuration has zero output under uniform tied attention.

The rank bounds and eigenvalue multiplicities are essential. If $V = 0$, every configuration is stationary. A nontrivial kernel admits every configuration whose tokens lie in that kernel. Repeated eigenspaces support continuous families obtained by rotating any one solution inside the eigenspace. Regular simplices and polygons give further examples only when their Gram matrix and multiplicities satisfy (16); an arbitrary spherical code need not be stationary. Distinct eigenspaces may occur in one equilibrium through solutions of (16), so choosing a single eigenvalue does not classify all stationary configurations.

3.2 An explicit mixed-eigenspace family

Let $Vu = au$ and $Vv = bv$ with $u \perp v$. Take m copies of u, k copies of $qu + \sqrt{1-q^2}v$, and k copies of $qu - \sqrt{1-q^2}v$, where $q \in (-1, 1)$. Under tied symmetric attention this configuration is stationary if and only if

$$q\left[(a-b)\mathrm{e}^{\beta[b+(a-b)q^2]} + (a+b)\mathrm{e}^{\beta[-b+(a+b)q^2]}\right] + a\frac{m}{k}\mathrm{e}^{\beta a q} = 0. \tag{20}$$

The equation follows by resolving the received output of either oblique group along the tangent direction $\sqrt{1-q^2}\,u - qv$. It gives stationary branches supported on two eigenspaces.

At $\beta = 0$, the classification is explicit even when V is nonsymmetric. Put $z = V \sum_j x_j$.

Theorem 3.4 (Uniform-attention equilibria)**.** *A configuration is stationary at $\beta = 0$ if and only if either*

1. *$z = 0$; or*
2. *$z \neq 0$, every token equals u or $-u$ for a real eigenvector u of V associated with a nonzero eigenvalue, and the signed token sum is nonzero.*

Proof. The vector field of token i is $(I - x_i x_i^{\mathsf{T}})z/n$. If $z \neq 0$, stationarity forces every x_i to be parallel to z, hence $x_i \in \{u, -u\}$ for $u = z/\|z\|$. Writing $\sum_i x_i = cu$ gives $z = cVu \parallel u$. Both c and the eigenvalue are nonzero. The converse and the case $z = 0$ follow by substitution. □

3.3 Tangent stability

Let $\delta x_i \perp x_i$ and define

$$\delta s_{ij} = \delta x_i^{\mathsf{T}} S x_j + x_i^{\mathsf{T}} S \delta x_j, \tag{21}$$

$$\delta a_{ij} = \beta a_{ij}\left(\delta s_{ij} - \sum_k a_{ik}\delta s_{ik}\right), \tag{22}$$

$$\delta A_i = \sum_j a_{ij} V \delta x_j + \sum_j \delta a_{ij} V x_j. \tag{23}$$

At an attention-only equilibrium $A_i = \lambda_i x_i$, the exact tangent linearization is

$$\delta\dot{x}_i = (I - x_i x_i^{\mathsf{T}})\delta A_i - \lambda_i \delta x_i. \tag{24}$$

Suppose a tied equilibrium $X^\star$ lies in a sum of selected eigenspaces of V. Let $K^\star = (a_{ij}(X^\star))$ and $\lambda_i^\star = (x_i^\star)^{\mathsf{T}} A_i(X^\star)$. For an eigenvector e_ℓ orthogonal to the selected subspace, with $Ve_\ell = \nu_\ell e_\ell$, perturbations $\delta x_i = z_i e_\ell$ form an invariant linear block

$$\dot{z} = L_\ell z, \qquad L_\ell = \nu_\ell K^\star - \mathrm{Diag}(\lambda_1^\star, \dots, \lambda_n^\star). \tag{25}$$

Indeed, the score has zero first variation in these directions. The in-subspace Jacobian together with one block (25) for every omitted eigenvalue gives the complete tangent test. Strictly negative spectral abscissa in all blocks implies hyperbolic attraction; a positive real part implies instability.

For consensus on an eigenvector of eigenvalue ν_p in the tied symmetric model, the strict local-attraction conditions of Kuehn and Yoon are

$$\nu_p > 0, \qquad \nu_k < \nu_p \quad (k \neq p). \tag{26}$$

For $\beta > 0$, the strict local-attraction conditions for a balanced bipolar split are

$$\nu_p > 0: \quad \nu_k < \nu_p \tanh(\beta\nu_p), \tag{27}$$

$$\nu_p < 0: \quad \nu_p < \nu_k < \nu_p \tanh(\beta\nu_p), \tag{28}$$

for every $k \neq p$. The second case selects the most negative eigendirection. These thresholds show why the sign pattern of the complete spectrum, rather than the top eigenvalue alone, controls bipolar stability [29]. At $\beta = 0$, a balanced split has zero mean and carries neutral split modes, so these strict-attraction criteria do not extend to uniform attention.

For $\beta > 0$, an unbalanced split with n_+ copies of u, n_- copies of $-u$, and $r = n_+/n_-$, put

$$c = \mathrm{e}^{2\beta\nu_p}, \qquad \sigma(c, r) = \frac{(c - r)(cr - 1)}{r(c^2 - 1)}.$$

If $\nu_p > 0$, local attraction requires

$$\nu_p > \frac{|\log r|}{2\beta}, \qquad \nu_k < \nu_p \sigma(c, r) \quad (k \neq p).$$

If $\nu_p < 0$, it requires

$$\nu_p < -\frac{|\log r|}{2\beta}, \qquad \nu_p < \nu_k < \nu_p \sigma(c, r) \quad (k \neq p).$$

At equality the linear test is nonhyperbolic. Setting $r = 1$ recovers (27)–(28).

For $S = V = \mathrm{Diag}(2, -3)$, $\beta = 1.5$, and three tokens, numerical root solving gives the angles $(0, 1.546889, 4.736296)$. The unrounded residual is 1.3×10^{-15} and the largest tangent real part is -0.9144. The three unnormalized received outputs have norms 40.27, 269.79, and 269.79 and are radial. This is a stable noncollinear clustering equilibrium under Altafini's terminology, rather than a zero-output polygonal equilibrium.

4 Fixed points, periodic orbits, and the continuous-depth limit

4.1 Finite-layer fixed points

Proposition 4.1 (Necessary and sufficient condition for a finite layer)**.** *A configuration X is a fixed point of (6) if and only if, for every token, there exist a unit intermediate point y_i and two strictly positive numbers r_i, s_i such that*

$$x_i + hA_i(X) = r_i y_i, \qquad y_i + hM(y_i) = s_i x_i. \tag{29}$$

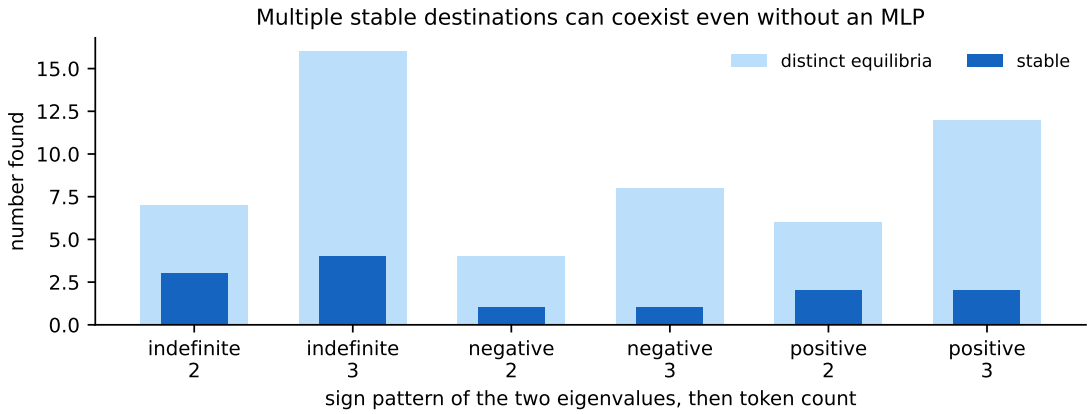


Figure 1: Numerically recovered stationary configurations for three spectra in the tied symmetric planar model. The plotted catalogue is not claimed complete. Theorem 3.2 gives the complete implicit stationary system; solving every branch of that transcendental system is a separate problem.

Proof. The first equality states exactly that the first normalization produces y_i; the second states that the second normalization returns to x_i. Positivity of r_i and s_i preserves rather than reverses direction. Conversely, normalizing both equalities recovers (6) and then $F_i(X) = x_i$. □

This form retains the softmax exponential inside A_i but removes the normalization divisions. It can therefore be used directly by an equation solver.

4.2 Finite-layer periodic orbits

For period p, introduce $X^k, Y^k \in (\mathbb{S}^{d-1})^n$ and $r_i^k, s_i^k > 0$ for $k = 0, \dots, p-1$. Write (29) at every step, replace the final X by the next state, and close the loop:

$$x_i^k + hA_i(X^k) = r_i^k y_i^k, \tag{30}$$

$$y_i^k + hM(y_i^k) = s_i^k x_i^{k+1}, \tag{31}$$

$$X^p = X^0. \tag{32}$$

To require primitive period, exclude returns before p. These equations therefore cover arbitrary periods and configurations, including nonpolygonal orbits.

4.3 Continuous-depth equation

For a unit vector x and a push z,

$$\mathcal{N}(x + hz) = x + h(I - xx^{\mathsf{T}})z + O(h^2).$$

Applying this identity twice to (6) gives

$$\boxed{\dot{x}_i = (I - x_i x_i^{\mathsf{T}})\{A_i(X) + M(x_i)\}.} \tag{33}$$

Liu and Herty identify this serial attention–MLP architecture as a Lie–Trotter splitting of the summed continuous equation [31]. The local calculation below records the order at which the two possible layer orderings first differ. Serial ordering appears only in the order-h^2 term. All continuous equilibria are therefore exactly the solutions of

$$\boxed{(I - x_i x_i^{\mathsf{T}})\{A_i(X) + M(x_i)\} = 0, \qquad i = 1, \dots, n.} \tag{34}$$

More explicitly, let the normalized attention and MLP submaps have local expansions

$$R_{A,h} = I + hf + h^2 r_A + O(h^3), \qquad R_{M,h} = I + hg + h^2 r_M + O(h^3),$$

where $f_i = P_{x_i}^{\perp} A_i$ and $g_i = P_{x_i}^{\perp} M(x_i)$. Then

$$R_{M,h} \circ R_{A,h} = I + h(f+g) + h^2\{r_A + r_M + Dg[f]\} + O(h^3), \tag{35}$$

$$R_{A,h} \circ R_{M,h} = I + h(f+g) + h^2\{r_A + r_M + Df[g]\} + O(h^3). \tag{36}$$

Their difference is $h^2(Dg[f] - Df[g]) + O(h^3)$. The serial block and a parallel continuous control share the first-order vector field, while finite layers retain this commutator correction.

At equilibrium the total push may be nonzero, provided that it is parallel to the token. Attention and the MLP may cancel in tangent directions even when neither subfield vanishes separately.

4.4 Stability

The stability calculation perturbs every token along tangent directions to its sphere. For a finite layer, spectral radius strictly below one for the restricted derivative implies local asymptotic stability, whereas an eigenvalue of modulus greater than one implies instability. For the continuous flow, negative spectral abscissa of the tangent Jacobian implies local asymptotic stability, whereas a positive real part implies instability. Eigenvalues on the unit circle or imaginary axis require higher-order analysis. The full tangent calculation includes perturbations that separate temporarily equal tokens.

5 Finite-layer census

The four families combine two choices for attention with two choices for the MLP.

Type	Attention	MLP
1	$S = V = V^{\mathsf{T}}$	potential
2	$S = V = V^{\mathsf{T}}$	general quadratic
3	untied S and V	potential
4	untied S and V	general quadratic

For types 3 and 4, we separated four subcases: symmetric S and V; symmetric S and general V; general S and symmetric V; and both general. MLP widths are 1, 2, 4, and 8; the main inventory uses one to four tokens.

The first broad sweep contains 28,672 models, with 96 starts per model and hence 2,752,512 trajectories. Families 1, 2, and 4 use 1,024 models at each token count; family 3 uses 4,096. Step size, sharpness, and the ratio of attention to MLP scale are continuous random variables. The five reported ranges for each quantity are post-hoc bins, not a factorial parameter grid. The ensemble and decision rules are specified in appendix A. The sweep is a finite sample from that law, not an enumeration of the continuous matrix space.

5.1 Stable fixed points

Direct equation solving finds the following mean numbers of stable fixed points.

Type	1 token	2 tokens	3 tokens	observed maximum
1	1.68	2.34	3.09	8
2	1.49	2.03	2.55	9
3	1.57	2.13	2.61	8
4	1.31	1.71	2.00	7

In the decoupled planar model with $\phi(\theta) = (g/3)\cos(3\theta)$ and $g > 0$, each token obeys $\dot{\theta} = -g\sin(3\theta)$. It has exactly three stable wells, hence exactly 3^n stable labeled product equilibria. All 3^n hyperbolic branches persist for sufficiently weak coupling by the implicit-function theorem. At coupling 0.03, seeded solves numerically continue 3, 9, 27, 81, and 243 distinct stable fixed points for one through five tokens. These counts are lower bounds at that finite coupling because the solves do not exclude additional roots.

5.2 Three- and four-layer cycles

The broad trajectory screen assigned period-three and period-four labels by numerical recurrence tests. Its raw counts are:

Type	screened p3 trajectories	p3 models	screened p4 trajectories	p4 models
1	321	8	201	15
2	2,039	34	1,823	28
3	4,490	75	4,507	113
4	1,621	22	1,860	41

These counts are screening labels, not 8,862 equation-level certificates. For each of the eight type–period pairs, one stored representative was subsequently refined by solving $F^p(X) = X$. All eight representatives have residual below 5×10^{-15}, no earlier return, and spectral radius below one for the derivative of F^p. The screen estimates how often its recurrence classifier fired under the sampled ensemble; the refinement establishes one stable primitive example in each cell.

The extended atlas contains numerically resolved primitive candidates at every period from 5 through 12. These occur across all four tested MLP widths and also under uniform attention ($\beta = 0$), but they are not all stable; in particular, the stored type-1 period-12 orbit is unstable. A general MLP alone can produce stable p3 and p4 cycles. A potential MLP alone produced no verified p3 in 65,536 trajectories, and its only p4 candidate reduced to a shorter period.

At finite step size, composing two normalized substeps can create a periodic orbit even when their infinitesimal vector fields are separately gradient-like. The finite-step normalization need not preserve either infinitesimal energy. In the continuous-depth limit, recurrence depends on circulation in the sum of the two vector fields.

6 Weak-layer limit

Suppose $F_h = I + hf + O(h^2)$. For a fixed integer period p,

$$F_h^p(x) = x + phf(x) + O(h^2). \tag{37}$$

A cycle that remains exactly three or four layers long as h tends to zero can only accumulate near a point where $f(x) = 0$. A genuine continuous cycle can survive, but the number of layers required for one revolution must grow as T/h, where T is its physical return time.

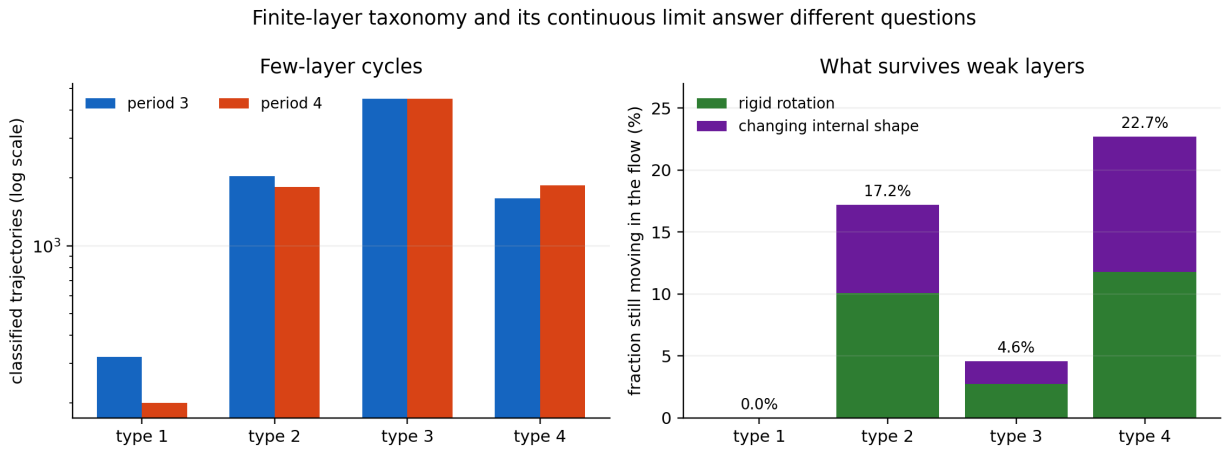


Figure 2: Left: few-layer cycles in the first sweep. Right: fraction of selected attractors that remains mobile after passing to the continuous equation. The panels have neither the same denominator nor the same question.

Fix the parameters, an initial condition, and $T < \infty$. Suppose that the two normalization inputs remain nonzero on a compact neighborhood of the corresponding flow segment, so that the local defect and its first derivative are uniformly bounded. Then, for sufficiently small h, the serial layer satisfies

$$\max_{0 \le k \le \lfloor T/h \rfloor} \left\| F_h^k(X) - \varphi_{kh}(X) \right\| = O(h), \tag{38}$$

where φ_t is the flow of (33). This convergence does not guarantee that two chaotic trajectories remain close forever. It guarantees that the continuous equation is the correct limit on every fixed time window. Trajectory distributions and basins are therefore checked separately.

6.1 Numerical protocol

The complete audit contains:

- 214,528 model configurations and 2,844,160 initial trajectories;
- 26,112 fresh, unselected controls in the 24 cells obtained by crossing the four types with 1, 2, 3, 4, 8, and 16 tokens;
- 23,789,394,642 model–layer updates, or 102,108,101,290 token–layer updates;
- 307,200,000 reference RK4 evaluations, or 1,740,800,000 token–field evaluations;
- steps down to $h/4096$, two nondyadic grids, physical times 2, 10, and 20, and Richardson controls that cancel the first-order error.

The error slope is 1 in every family. After Richardson cancellation it is 2.018 over the main population. At the finest step, the median error falls from 2.63×10^{-4} to 3.39×10^{-7}. These measurements support the first-order expansion $F_h = I + hf + h^2 g + \cdots$ on the sampled parameter sets.

6.2 Continuous-flow classification

The 1,908 selected attractors are replayed directly under (33).

Type	replayed	fixed	moving	rigid rotation	changing internal shape
1	286	286	0	0	0
2	378	313	65	38	27
3	768	732	36	21	15
4	476	368	108	55	53
Total	1,908	1,699	209	114	95

Thus 89.0% of the complex finite-step attractors selected become fixed points, while 11.0% remain mobile over the main window. This rate is conditional on selecting a complex attractor; it is not their frequency among all random matrices. The table uses the strengthened replay with two perturb-and-relax stages. One numerical synchronization lock was reclassified as moving after a longer recheck, and one moving type-4 record was reclassified from rigid to internal motion. Before these audited corrections the same window gave 1,700 fixed, 208 moving, and 93 internal records.

For 8 and 16 tokens, 581 candidates are continued. At step $h/4096$, the raw count is 118 moving states. Four apparently fixed states have an unstable direction hidden by numerically identical tokens; after correction, the finite map and continuous flow both give 122 moving states and agree case by case in 575 of 581 instances. The six remaining differences select coexisting basins.

7 Recurrent dynamics in the continuous flow

7.1 An exact type-3 limit cycle

Let $d = 2$, write $x(\theta) = (\cos\theta, \sin\theta)$, and let J denote rotation by $\pi/2$. Take uniform attention, $S = 0$, value transport $V = aI + \omega J$ with $a > 0$, and the potential MLP

$$\phi(x) = \frac{g}{3}\,\mathrm{Re}(x_1 + \mathrm{i}x_2)^3. \tag{39}$$

Proposition 7.1 (Orbitally stable active rotator)**.** *If $|\omega| > |g|$, the synchronized manifold $x_1 = \cdots = x_n = x(\theta)$ is an exact periodic orbit of (33). Its phase satisfies*

$$\dot\theta = \omega - g\sin(3\theta), \qquad T = \frac{2\pi}{\sqrt{\omega^2 - g^2}}. \tag{40}$$

The collective Floquet exponent is zero and all $n-1$ transverse exponents equal $-a$. The orbit is therefore locally orbitally asymptotically stable.

Proof. At synchrony the uniform attention output is $ax(\theta) + \omega Jx(\theta)$. Its radial part disappears under projection, while the tangential part of $\nabla\phi$ is $-g\sin(3\theta)$, which gives (40). The standard integral over one rotation gives the stated period. Write $\theta_i = \theta + \eta_i$. To first order,

$$\dot\eta_i = a(\bar\eta - \eta_i) - 3g\cos(3\theta)\eta_i.$$

The collective component is the phase variation. For every zero-mean transverse component the instantaneous rate is $-a - 3g\cos(3\theta)$. The second term has zero period average because

$$\int_0^{2\pi} \frac{\cos(3\theta)}{\omega - g\sin(3\theta)}\,d\theta = 0.$$

Hence every transverse Floquet exponent is $-a$. □

For the experimental values $a = 1.2$, $\omega = 0.8$, and $g = 0.25$, the exact mean angular speed is $\sqrt{\omega^2 - g^2} = 0.759934\ldots$. The 128-start simulations measure the basin reached by the serial discretization; they are not needed for the existence or local stability of the continuous orbit.

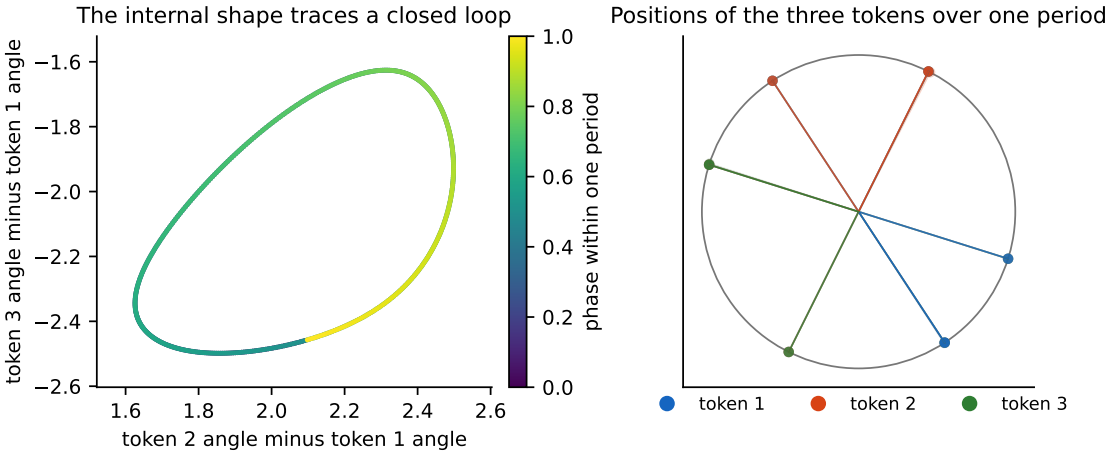


Figure 3: Numerically resolved attracting internal cycle under attention alone. The relative-angle curve is nonconstant, and seven phases show the changing three-token geometry.

7.2 Stable cycles

Four recurrent motions are numerically resolved in the block with an MLP: one p3 and one p4 seed from type 2, followed by one p3 and one p4 seed from type 4. At the deepest dyadic steps, their full labeled-state return times agree with RK4: respectively 16.632 versus 16.660, 9.331 versus 9.340, 9.389 versus 9.400, and 6.4395 versus 6.4400. On nondyadic grids, fitted layer-count exponents are 1.0016, 1.0029, 0.9992, and 0.9950. The second fit uses a Gram/shape return; it does not by itself validate a full-state period. Perturbed trajectories return to the same recurrent motion in every robustness run.

The type-4 p3 case is especially informative: attention alone retains the cycle while the MLP alone stops. With the MLP removed, its stored parameters are

$$\begin{aligned} S &= \begin{pmatrix} 1.112495 & 1.768642 \\ -2.540753 & 3.294286 \end{pmatrix}, \\ V_0 &= \begin{pmatrix} -1.809502 & 1.419182 \\ -1.827949 & -3.611608 \end{pmatrix}, \\ \beta &= 1.110063, \qquad \rho = 0.551488. \end{aligned} \tag{41}$$

The audit uses normalized depth τ and integrates $dX/d\tau = \rho f_{S,V_0,\beta}(X)$. This is the flow (7) after the time change $t = \rho\tau$, or equivalently after replacing V_0 by ρV_0. Independent RK4 integration gives return time $T_\tau = 9.120$ and hence $T_t = 5.02957$ in the convention of (7). The pairwise inner products range from -0.8002 to -0.0551, excluding a rigid rotation. Over 2,000 units of τ, the Benettin–QR estimate of the Lyapunov spectrum per unit τ is

$$(-0.00273, -0.45676, -0.45937), \tag{42}$$

or $(-0.00495, -0.82823, -0.83296)$ per unit t. The first rate approximates the neutral phase direction. For 128 Gaussian angular perturbations of standard deviation 0.08, the maximum phase-minimized distance after 36 units of τ is 0.003673 radians, compared with a maximum initial distance of 0.142421. Across seven dyadic and nondyadic layer ratios, the fitted slope of layers per revolution against inverse layer strength is 1.012. These tests are consistent with convergence to an attracting continuous-depth periodic orbit; they are not a computer-assisted existence proof.

In the other three numerically resolved recurrent examples, the MLP participates in the motion.

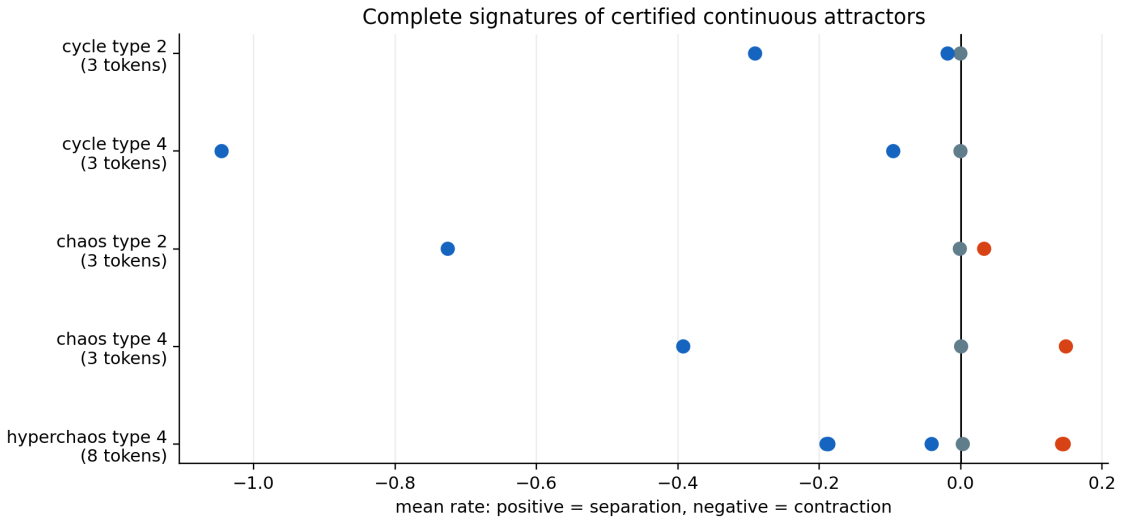


Figure 4: Complete rates for cycles and chaotic-attractor candidates. An orange point to the right of zero means that a small difference grows; a blue point to the left means it contracts; a gray point near zero follows the phase of the motion.

7.3 Three-token chaotic-attractor candidates

Here a motion is called chaotic only after four checks: one perturbation direction grows on average; one direction remains neutral because it follows time along the trajectory; the remaining directions contract; and the sum of rates agrees with the independently computed mean divergence.

The three-token type-4 case has rates

$$(0.1489,\ 0.0007,\ -0.3923).$$

The leading rate remains positive in all 24 perturbations and for three RK4 steps; at step 0.005 it equals 0.1584. All 512 tested uniform starts reach an internal motion with positive rate within the stated observation window. Attention alone stops, the MLP alone produces a regular rotation, and their interaction gives strong numerical evidence of chaos.

A second type-2 example has rates

$$(0.0333,\ -0.0011,\ -0.7253).$$

Here attention alone and the MLP alone both stop. Among 512 starts, 449 exceed the strict 0.03 threshold and the other 63 remain positive between 0.022 and 0.03.

7.4 An eight-token hyperchaotic-attractor candidate

One type-4 example has three independent separating directions:

$$(0.1458, 0.1455, 0.1435, 0.0031, -0.0409, -0.1871, -0.1872, -0.1896).$$

A replication with a step twice as small retains three positive rates. Four tokens remain synchronized during this motion; the trajectory alternates mainly between $4+4$ and $4+3+1$ partitions. It is a chimera: one part of the group remains ordered while group phases evolve unpredictably. Attention weights vary very little, whereas token geometry varies strongly.

Tomihari and Karakida previously used state Jacobians and finite-time Lyapunov exponents for normalized recurrent self-attention, including nonsymmetric and multihead layers [44]. They show

that normalization can keep the Jacobian spectrum near criticality and report slightly positive maximum finite-time exponents, with larger values in some multihead models. Their horizon is $T = 16$ and their quantity does not establish an asymptotic chaotic attractor. Our long-horizon spectra, divergence checks, and perturbation ensembles address that numerical distinction, while still falling short of a computer-assisted proof. The exact active rotator above supplies an existence theorem for a periodic orbit, not for the chaotic examples.

7.5 Recurrent motion under uniform attention

At $\beta = 0$, every weight always equals $1/n$. Nevertheless, a four-token type-3 case retains intermittent motion with positive leading rate. The source is the antisymmetric part of V. Replacing it by its symmetric part, removing attention, or removing the MLP causes convergence to a fixed point. In an audit of 512 starts, the same weight set has fixed points, recurrent motions, and positive-rate trajectories: initialization selects the basin.

On the circle, this case rewrites exactly as a system of coupled oscillators. The antisymmetric part of V supplies the tendency to rotate, while the potential MLP supplies regions where motion slows. Uniform attention is therefore not an absence of interaction; it is coupling through the global mean.

No robust quasiperiodic torus was found in the audited sample. The observed classes include fixed points, stable spirals, periodic orbits, and trajectories with numerical signatures of chaos or hyperchaos. This is not an exhaustive classification of a high-dimensional continuous system.

8 Potential structure and circulation

When $\beta = 0$, $V = V^{\mathsf{T}}$, and the MLP is potential, the function

$$\Phi(X) = \frac{1}{2n} \sum_{i,j} x_i^{\mathsf{T}} V x_j + \sum_i \phi(x_i) \tag{43}$$

increases along trajectories:

$$\frac{d\Phi}{dt} = \sum_i \left\| (I - x_i x_i^{\mathsf{T}}) \nabla_{x_i} \Phi \right\|^2 \geq 0.$$

A nonconstant loop would return to the same value after a revolution despite having strictly increased, which is impossible. All 146 numerical trajectories in this exact-potential control were classified as converged on the stated observation window.

Row-wise softmax normalization can break this simple structure even when $S = V = V^{\mathsf{T}}$. Among 1,699 audited equilibria, 28 are stable spirals: one in type 1, 19 in type 3, and eight in type 4; none appears in the exact $\beta = 0$ control. The single type-1 spiral shows that two symmetric subfields with different mobilities do not guarantee a single Euclidean energy.

For unnormalized multihead attention, properly aligned symmetric heads add into a common energy. With separate normalizations, this conclusion needs additional conditions because each head multiplies its push by its own denominator. Massucco and collaborators give an energy with a nonlocal mobility for a specified tied time-dependent setting [34]. Rodríguez Abella, Silvestre, and Tabuada prove consensus for bounded, uniformly continuous time-dependent score matrices and any number of independently normalized heads when every transported output is the identity and the tokens begin in a common hemisphere [1]. Our examples leave those hypotheses through general output matrices, untied transport, or token configurations spanning more than one hemisphere.

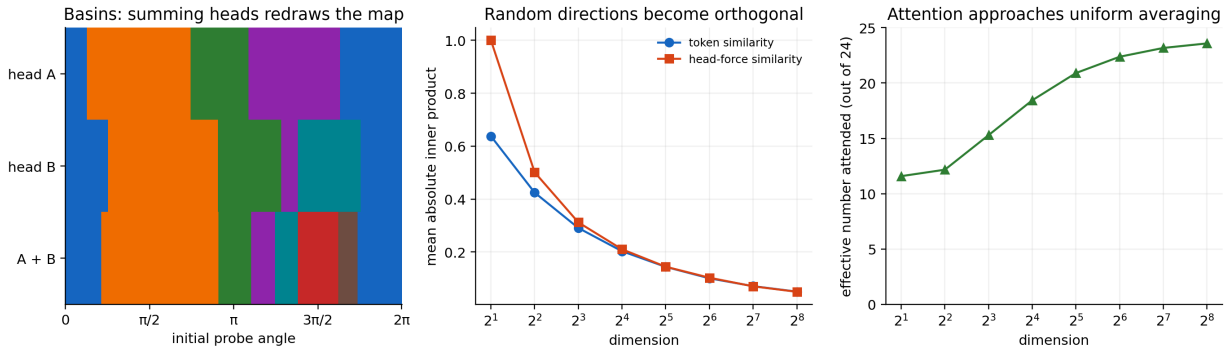


Figure 5: Left: probe destination as a function of its initial angle for head A, head B, and their sum. Center: mean absolute inner products between tokens and between head forces. Right: effective number of attended tokens, defined as the exponential of the row-attention entropy. The experiment uses 24 tokens, six random heads, and 96 repetitions per dimension.

9 Multihead equilibria and basin geometry

For L heads, the push is

$$A_i^{\text{multi}}(X) = \sum_{\ell=1}^{L} \omega_\ell \sum_j a_{ij}^{(\ell)}(X) C_\ell x_j. \tag{44}$$

The score matrix of head ℓ determines its weights; C_ℓ includes its value matrix and output recombination. The destination satisfies

$$(I - x_i x_i^{\mathsf{T}})\{A_i^{\text{multi}}(X) + M(x_i)\} = 0.$$

Each head need not vanish separately. Two heads can cancel along the sphere and create a destination that exists for neither head alone.

9.1 Conditional basin maps

We fix five tokens on the circle and allow a sixth probe to move. With two matrices having the same spectrum $(2, -3)$ but axes rotated by 60 degrees, head A has four destinations, head B has five, and their sum has seven. On the sphere the counts are four, four, and five. The sum cuts boundaries, moves destinations, and creates small basins. Two very close initial points can therefore end in distant groups if they lie on opposite sides of a boundary.

9.2 Settling-time maps and fractal diagnostics

Lai, Bao, Quinn, and Gilpin recently reported fractal settling-time maps, called basin plots in their paper, for trained recurrent reasoning models [30]. Their construction fixes a model and a task, chooses a random two-dimensional plane in the full latent space, and colors each initial condition by the number of recurrent loops until the decoded output stops changing. Consequently, adjacent colors may represent different transient durations on routes to the same final answer. Across their population plots, basin entropy increases with the mean loop count on the same slice; examples ordered by task-specific difficulty also follow this trend. They attribute the long transients to trajectories lingering near unstable, nearly correct states. These correlations do not constitute a complexity lower bound. They repeat spatial zooms and estimate uncertainty exponents. For an Equilibrium Reasoner on Sudoku, the reported values are 0.249, 0.243, and 0.216 across the base, tenfold, and hundredfold views; a Parcae example on Countdown and a Fixed-Point Reasoning Model example on mazes give substantially larger values, roughly 0.85–0.91. Their finite latent

instability statistic compares neighboring decoded trajectories at finite separation. In a separate toy training model they report finite-time Lyapunov exponents and Jacobian eigendirections.

Their stopping rule is applied to the decoded answer. A latent trajectory may still move after its discrete decoding has stopped changing. Our reduced model has no task decoder, so the experiment below uses the norm of the token-state vector field and records continuous settling time. It is an analogue of their observable, not an identical measurement.

Their training experiment supplies a second trend. An eight-variable looped Transformer for modular linear systems initially has several stable fixed points that decode to incorrect solutions. In four closely spaced reported checkpoints near 156,000 training steps, these states lose stability, the correct solution becomes the common destination, basin entropy rises, and unstable Jacobian directions appear. Positive finite-time exponents occur in variables requiring multistep elimination, not in variables solved by direct substitution. These measurements show a coincident inference-space bifurcation and gain in task accuracy for that trained model. They do not establish that computational hardness forces the same transition in every architecture.

Their population statistics use 200×200 slices, pixel separations one through five, at least 300 valid slices per model–task pairing, and retain only slices for which at least 90% of starts reach the final solution and at most 1% hit the iteration cap. The supplemental zooms produce stable or drifting fitted exponents, which the authors interpret respectively as scale-stable and "slim" fractals. A repeated co-moving zoom is itself subject to the smooth-saddle null derived below. This distinction is relevant for dissipative dynamics: convergence to one final state does not force the settling-time field to have a smooth boundary [35, 16].

The relation between constraint solving and transient chaos predates recurrent reasoners. Ercsey-Ravasz and Toroczkai construct a continuous-time SAT solver whose solution clusters are attractors and whose basin boundaries become fractal beyond a constraint-density threshold [19]. Daza et al. introduced basin entropy to quantify final-state uncertainty and gave a sufficient entropy criterion for fractal basin boundaries [17]. These results justify the dynamical diagnostics; they do not turn an arbitrary recurrent system with an intricate basin plot into a solver for a hard problem. Such a conclusion for our model would require an explicit map from problem instances to S, V, M, a decoding map from attractors to solutions, and a cost analysis that includes precision and transient time. Daza et al.'s sufficient criterion applies to genuine terminal-basin labels; it does not certify a map colored only by settling time.

Our plot in figure 5 measures a different object. It varies one probe angle over 1,440 points and colors the terminal attractor identity. Heads A and B and their sum have respectively $(4, 5, 7)$ destinations and $(4, 5, 7)$ label transitions around the circle. This establishes multistability and head-induced partition refinement. It contains no repeated zoom and gives no evidence of a fractal boundary.

The small-step continuation audit also measures a different exponent. For two potential landscapes, the fraction of initial states whose token partition differs between a finite layer and the continuous flow scales approximately as $h^{0.946}$ and $h^{0.958}$ over the main range. Here h is layer strength, not a perturbation of the initial condition. The values are consistent with an $O(h)$ displacement of a smooth codimension-one separatrix and cannot be interpreted as uncertainty exponents.

A simple saddle and a chaotic saddle make different predictions. Near a hyperbolic saddle with unstable rate $\lambda_u > 0$, local linearization gives the settling-time singularity

$$T(x) \simeq \frac{1}{\lambda_u} \log \frac{C}{\operatorname{dist}(x, W^s)}, \tag{45}$$

where W^s is its stable manifold. A smooth hyperbolic intersection of W^s with a fixed planar slice has uncertainty exponent one in the asymptotic small-separation limit even though adjacent pixels can have very different settling times. For the standard terminal-basin uncertainty test, an exponent below one under the usual scaling assumptions indicates a boundary of dimension

greater than one; recurrent folding by a chaotic saddle is one mechanism that produces it [23]. Long transients and a visually intricate finite grid are therefore insufficient on their own.

A related test for our dynamics records both terminal attractor and settling time on a two-dimensional state slice. Write $\vartheta = (\theta_1, \theta_2)$ for the two slice coordinates. Repeated spatial magnifications can then estimate, for a stated time threshold $\delta > 0$,

$$f_\delta(\varepsilon) = \Pr\big(|T(\vartheta) - T(\vartheta')| \geq \delta \mid \|\vartheta - \vartheta'\|_2 = \varepsilon\big) \asymp \varepsilon^{\alpha_\delta}. \tag{46}$$

A pair is uncertain here only when its continuous settling times differ by at least δ; this is a coarsened analogue of comparing unequal integer loop counts. A fitted exponent below one under repeated co-moving zooms is not sufficient. For the smooth-saddle law (45), let $d > 0$ denote the normal distance on one side and compare d with $d + \varepsilon$. Then

$$|T(d+\varepsilon) - T(d)| \geq \delta \quad \Longleftrightarrow \quad d \leq \frac{\varepsilon}{e^{\lambda_u \delta} - 1}. \tag{47}$$

The uncertain fraction is asymptotically proportional to ε on a fixed window, but it saturates when the right-hand side is comparable to the window width. Shrinking the window and the pixel spacing by the same zoom factor preserves this dimensionless ratio and can preserve a low finite-scale fitted slope for a perfectly smooth stable manifold. Evidence for a fractal boundary therefore requires fixed-window resolution refinement beyond this saturation regime, in addition to integration-step and horizon controls. A positive finite-time tangent rate during one saddle passage is also compatible with a smooth boundary; recurrent folding or an invariant chaotic saddle requires separate evidence. Terminal-attractor labels require a separate basin-uncertainty calculation. The tied symmetric continuous attention flow cannot contain a nonstationary chaotic invariant set because of (18); a geometrically complicated field of slow settling near stationary saddles remains possible. Untied heads, a general MLP, or a finite recurrent map also admit recurrent saddles. The four-token untied example with 147 fixed and 365 moving or unresolved outcomes among 512 starts is a candidate for this test; its existing sample does not establish fractality.

A separate three-token type-2 candidate used for the atlas below also permits an exact one-dimensional reduction. On the synchronized manifold, write $x(\theta) = (\cos\theta, \sin\theta)$ and $x(\theta)^\perp = (-\sin\theta, \cos\theta)$, and let $\rho = 1.4468864491008047$ be the source record's overall depth scale. Its angular field is

$$\dot\theta = g(\theta, \gamma) = \rho x(\theta)^{\perp\mathsf{T}}\{V x(\theta) + \gamma M(x(\theta))\}. \tag{48}$$

A numerical evaluation using 60-digit arithmetic on the stored decimal coefficients gives a saddle–node at

$$(\theta_c, \gamma_c) = (1.1095854010814425,\ 1.4347884764108496),$$
$$g_\gamma = 0.2655849737, \qquad g_{\theta\theta} = -9.5222584300.$$

This evaluation is neither symbolic nor interval-certified. With $u = \theta - \theta_c$ and $\mu = \gamma - \gamma_c$, Taylor expansion gives

$$g(\theta_c + u, \gamma_c + \mu) = 0.2655849737\mu - 4.7611292150u^2 + O(|u|^3 + |\mu u| + \mu^2). \tag{49}$$

At the same numerical point, central differences of the full three-angle field give the spectrum

$$(0, -0.05706479, -0.05706479).$$

The zero collective direction and the two strictly contracting transverse directions are consistent with a generic saddle–node of the full field. This remains a floating-point calculation rather than a computer-assisted proof. For $\mu > 0$, the predicted separation of the two synchronized roots is $0.4723639672\sqrt{\mu} + O(\mu)$, and the magnitudes of their collective eigenvalues are $2.2489858845\sqrt{\mu} + O(\mu)$. At $\gamma = 1.4348$, the stable collective eigenvalue is -0.00763411, corresponding to a linear

relaxation time of 130.99; its other two angular eigenvalues are both -0.06530978. The companion saddle has spectrum $(0.00763495, -0.04881867, -0.04881867)$. This slow scale can create sharp finite-resolution changes in settling time without a fractal boundary.

The fold itself supplies a stronger null for the settling-time statistic. At $\mu = 0$, the leading normal equation on its repelling side is

$$\dot{u} = -au^2, \qquad T(u) \sim \frac{1}{a|u|}, \qquad a = 4.7611292150. \tag{50}$$

Put $s = |u| > 0$ on this side of the fold and compare s with $s + \varepsilon$. For the leading normal form,

$$|T(s+\varepsilon) - T(s)| \geq \delta \quad \Longleftrightarrow \quad s(s+\varepsilon) \leq \frac{\varepsilon}{a\delta},$$
$$0 < s \leq s_*(\varepsilon) := \frac{\sqrt{\varepsilon^2 + 4\varepsilon/(a\delta)} - \varepsilon}{2} \sim \sqrt{\frac{\varepsilon}{a\delta}}.$$

For a sampling density that is continuous and positive at the fold, the uncertain fraction is therefore $f_\delta(\varepsilon) = \Theta(\varepsilon^{1/2})$. The exponent is one half although the singular set is smooth. When $\mu > 0$, the hyperbolic regime is recovered only within a distance much smaller than

$$r = \sqrt{g_\gamma \mu / a}$$

from either root. For the atlas value $\gamma = 1.4348$, $r = 8.0175 \times 10^{-4}$ and the root separation is $2r = 1.6035 \times 10^{-3}$. The deepest-window initial-angle separations range from 1.9175×10^{-4} to 3.0680×10^{-3}. The preceding flow distorts these separations before the bottleneck, so their comparison with r only locates a possible crossover.

The numerical atlas contains 30,656 integrated initial conditions. The four 64×64 primary windows contain respectively 4,096, 4,096, 4,088, and 4,010 trajectories satisfying the finite-horizon speed-and-tail criterion. A post-hoc comparison with the synchronized roots rejects 0, 1, 2, and 17 additional endpoints because they are closer to the saddle than to the stable root. The remaining endpoints are consistent with one stable consensus. The three clusters obtained by applying a 10^{-3} tolerance directly to all finite-time endpoints record positions along the weak synchronized direction and are not evidence for three attractors.

For $\delta = 2$, the root-audited finite-scale slopes across magnifications $1, 8, 64, 512$ are

$$1.025 \pm 0.077, \qquad 0.855 \pm 0.057, \qquad 0.715 \pm 0.022, \qquad 0.663 \pm 0.070.$$

The quoted uncertainties are ordinary least-squares standard errors from five correlated pixel offsets, not independent-sample confidence intervals. At the deepest zoom the slope changes from 0.480 for $\delta = 1$ to 1.262 for $\delta = 10$. Across the speed-based controls, halving the integration step changes any fitted slope by at most 0.0030, whereas changing the grid from 64^2 to 32^2 changes a slope by as much as 0.417 and extending the deepest control from time 220 to 300 changes it by as much as 0.115. The exponent has therefore not converged with respect to the spatial, time-threshold, and horizon choices.

The geometry at the two finest windows is simpler than the fitted slopes suggest. At magnification 512, all 103 pixels excluded by the combined speed-and-root-distance audit form one contiguous strip: every column has one interval and two boundary crossings. Affine fits to its lower and upper pixel edges have root-mean-square residuals 0.282 and 0.288 pixel, and the minimum-time ridge has residual 0.284 pixel. No repeated folding is resolved. The largest finite-time tangent exponent is -0.00892 along the longest retained trajectory and -0.09724 along a typical one; at $\gamma = 1.43475$, just below the fold, the audited trajectory gives -0.04955. These values persist under halving the step and reducing the perturbation from 10^{-7} to 10^{-8}. Positive local stretching intervals occur, but the full audited intervals are contracting.

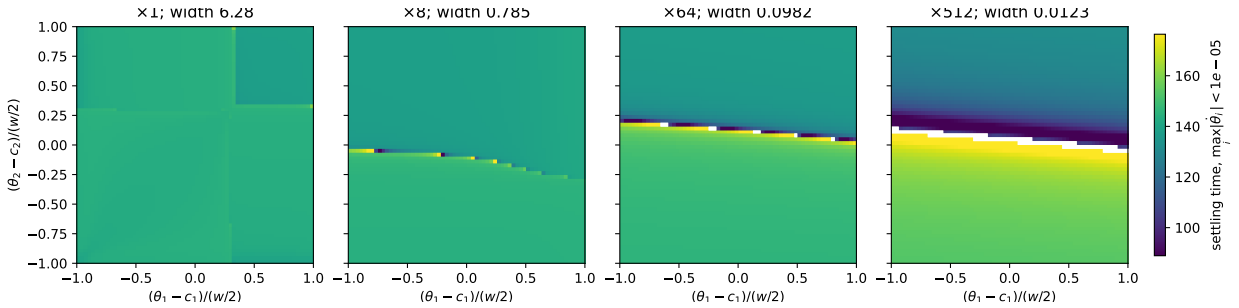
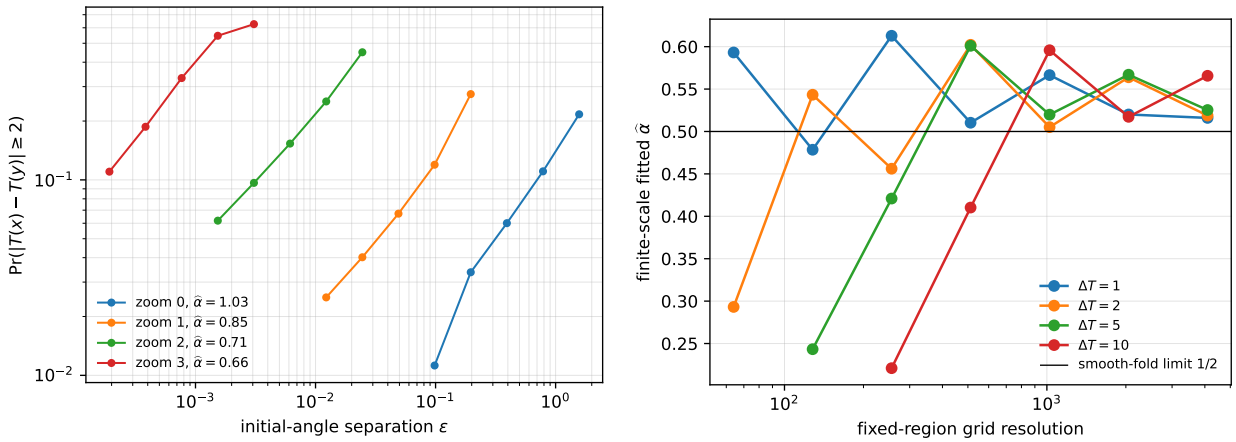


Figure 6: Finite-horizon settling-time atlas at four adaptively selected magnifications. The first two initial angles vary and the third is fixed; axes are normalized separately in each window. White pixels fail the speed-and-tail criterion by time 220. Beyond the base view, the transition remains a single strip at the displayed resolutions.



(a) Measured statistic at $\delta = 2$.

(b) Smooth critical-fold null.

Figure 7: Left: finite-scale power-law fits drift from 1.03 to 0.66 under co-moving zoom. Right: the nonfractal normal form $\dot{u} = -4.761129215u^2$ approaches the analytic exponent 1/2 under fixed-window resolution refinement. These regressions describe sampled scales; they are not fractal-dimension estimates.

The parameter scan agrees with the scalar bifurcation. None of the 576 base-grid trajectories satisfies the speed-and-tail criterion by time 220 at $\gamma \in \{1.4346, 1.4347, 1.43475\}$, while all 576 satisfy the speed-and-tail criterion at each tested $\gamma \geq 1.4348$. Since the atlas parameter exceeds γ_c by only 1.1524×10^{-5}, critical slowing is expected. In the analytic hyperbolic null, a 64^2 co-moving grid gives a fitted slope 0.0046 for $\delta = 2$ and the same value at every zoom; fixed-window refinement gives 0.996 at resolution 2,048 and 1.003 at 4,096. The fold null instead approaches one half, giving 0.564 and 0.519 at those resolutions. Low finite-scale slopes and their persistence under co-moving zooms therefore occur for smooth sets in both relevant local models.

The experiment therefore does not exhibit an analogous fractal settling-time map in this parameter slice. It reproduces the visual fact that nearby starts can have substantially different stopping times. Their retained quiet finite-horizon endpoints are consistent with approach to the same stable consensus. At the tested scales, the observed structure is quantitatively consistent with critical slowing near a smooth saddle–node; neither the map geometry nor the three-trajectory tangent audit provides evidence of a chaotic saddle.

9.3 A rotating cluster from two head outputs

Consider two planar heads with the same score $S = I$ and outputs $C_1 = I$ and $C_2 = \omega J$, where J rotates a vector by $\pi/2$.

Proposition 9.1 (Exact multihead rotating consensus)**.** *For $\omega \neq 0$, the synchronized manifold $x_1 = \cdots = x_n = x(\theta)$ is a periodic orbit with $\dot{\theta} = \omega$ and period $2\pi/|\omega|$. The collective exponent is zero and all transverse exponents are -1.*

Proof. At synchrony every row of each attention matrix is uniform, and the summed output is $x(\theta) + \omega J x(\theta)$. Projection removes the first term. If $\theta_i = \theta + \eta_i$, then $x_i^{\mathsf{T}} x_j = 1 + O((\eta_i - \eta_j)^2)$, so the attention weights have no first-order variation. The linearized phase equation is $\dot{\eta}_i = \bar{\eta} - \eta_i$. It has one zero collective rate and rate -1 on the zero-mean subspace. □

With $\omega = 0.8$ and 64 tokens, the simulation measures synchronization increasing from 0.471 to 1.000 and angular speed 0.800000. This probes the larger basin of the exact orbit. Its Gram matrix is constant: the score kernel records no motion although the transported values rotate. Since the two heads share the same score, their sum is algebraically one head with effective output $I + \omega J$; the example proves stable rotation of summed outputs, not an irreducible effect of distinct attention kernels.

10 High dimension: orthogonality without decoupling

For two random unit directions in dimension d,

$$\mathbb{E}|x^{\mathsf{T}} y| \simeq \sqrt{\frac{2}{\pi d}}.$$

With 24 tokens, six random heads, and 96 repetitions per dimension, mean absolute token similarity falls from 0.638 in dimension 2 to 0.050 in dimension 256. Mean similarity between head forces falls from 1.000 to 0.049. Independent heads therefore combine almost like perpendicular directions.

Their scores also become similar. For fixed $\beta = 3$, the effective number of attended tokens out of 24 rises from 11.6 to 23.6. The dynamics does not split into 24 independent systems; it retains an almost uniform global mean.

If $S = I$, the self-score remains one while cross-scores tend to zero. The limiting self-weight is

$$a_{ii} \longrightarrow \frac{\mathrm{e}^{\beta}}{\mathrm{e}^{\beta} + n - 1}. \tag{51}$$

Thus β must be appreciably larger than $\log n$ for genuine self-decoupling. In the experiment's Wigner ensemble $S = (G + G^{\mathsf{T}})/\sqrt{2d}$, cross-scores are $O(d^{-1/2})$; retaining nontrivial logits therefore requires $\beta/\sqrt{d}$ to remain of constant order, rather than tend to zero. This scaling is tied to that ensemble.

In a practical Transformer, division by $\sqrt{d_{\text{head}}}$ and learned query and key norms are specifically chosen to keep logits of order one. Raw orthogonality of embeddings is therefore insufficient to predict decoupling in a trained network.

Gao, Yang, and Chen obtain the complementary high-dimensional regime [20]. Their tokens have norm $\sqrt{d}$, self-edges are masked, $S = V = I$, and the logits use the standard $1/\sqrt{d}$ factor. If within-cluster overlaps exceed every cross-cluster overlap by a fixed gap Δ_a, the leakage from a cluster of size n_a is exponentially small when

$$\beta\sqrt{d}\,\Delta_a - \log\frac{n - n_a}{n_a - 1} \longrightarrow +\infty. \tag{52}$$

Their limiting clustered manifolds are normally attracting in internal directions; collective rotations are neutral, and finite systems can coarsen on longer scales. Our fixed-unit-norm, fixed-β experiment instead sends cross-logit differences to zero and therefore approaches uniform weights. The conclusions differ because the logit gap is $O(\sqrt{d})$ in (52) and $O(d^{-1/2})$ in our Wigner-score control.

In the low-temperature mean-field limit, Alcalde, Bungert, Riedl, and Roith prove a finite-time concentration estimate under invertibility of S and symmetry plus a spectral gap for VS^{T} [4]. Up to times of order $\log \beta$, the selected subspace is controlled by the top eigenspace of VS^{T}; their later numerical phase is controlled by the spectrum of V. This gives one rigorous setting in which score–value selection and terminal value transport occur on distinct time scales. The symmetry of VS^{T} excludes the circulating examples studied below.

11 Many-token limit

Define the empirical measure

$$\mu_t^n = \frac{1}{n}\sum_{i=1}^{n} \delta_{x_i(t)}.$$

For a positive kernel g and normalized attention, the velocity at x is

$$v_g[\mu](x) = P_x^{\perp} \frac{\int g(x^{\mathsf{T}}Sy)Vy\, d\mu(y)}{\int g(x^{\mathsf{T}}Sy)\, d\mu(y)} + P_x^{\perp} M(x). \tag{53}$$

Assume $g \in C^1$ is strictly positive on $[-\|S\|_{\mathrm{op}}, \|S\|_{\mathrm{op}}]$ and that M is Lipschitz on the sphere. The denominator is then uniformly bounded away from zero, and $v_g[\mu](x)$ is Lipschitz jointly in x and in μ for the 1-Wasserstein distance. The continuum equation is

$$\partial_t \mu_t + \mathrm{div}_{\mathbb{S}}(\mu_t v_g[\mu_t]) = 0. \tag{54}$$

The empirical measure of the n-token ODE is an exact atomic weak solution of this equation. The characteristic stability estimate gives, on every fixed time interval,

$$W_1(\mu_t^n, \mu_t) \le \mathrm{e}^{Ct} W_1(\mu_0^n, \mu_0), \tag{55}$$

where C depends on the fixed kernel, matrices, MLP, and interval. Thus convergence of the initial empirical measures implies the many-token limit. This is the standard Dobrushin argument for Lipschitz mean-field transport [18]; Transformer-specific transport formulations are developed by Rigollet and by Castin, Ablin, Carrillo, and Peyré [41, 14].

In our static quadrature control, a three-bump density is sampled and compared with an 8,192-point reference. The velocity error decays as $n^{-0.494}$, consistent with the $n^{-1/2}$ Monte Carlo fluctuation: 0.289 at 16 tokens, 0.146 at 64, 0.071 at 256, and 0.038 at 1,024. This measured exponent concerns one field evaluation; the dynamical convergence statement is (55).

The deterministic drift limit used here scales both residual branches as h and keeps their weights fixed over depth. Agazzi, Bruno, Mosig García, Saviozzi, and Romito analyze the same serial order—attention, sphere normalization, one-hidden- layer MLP, sphere normalization—with fresh independent MLP weights at every layer and MLP residual scale $\sqrt{h}$ [2]. They prove convergence to a common-noise interacting diffusion, the associated stochastic transport equation, propagation of chaos, and synchronization under a coercivity condition on the noise. Thus random MLP weights in that scaling do not reduce to the deterministic field M in (53); the common Brownian field can itself drive synchronization.

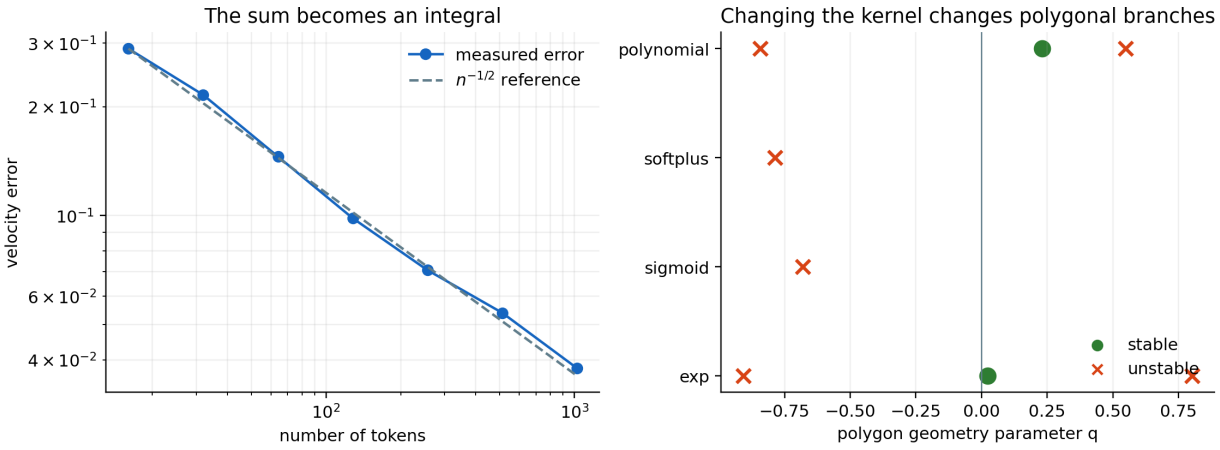


Figure 8: Left: convergence of the token sum to the integral. Right: polygonal positions found for four kernels; a green circle denotes a stable branch and an orange cross an unstable one.

11.1 Polygons become masses

A distribution concentrated at q positions has the form

$$\mu = \sum_{a=1}^{q} p_a \delta_{w_a}, \qquad p_a > 0, \quad \sum_a p_a = 1.$$

Set

$$Z_a = \sum_b p_b g(w_a^\mathsf{T} S w_b).$$

It is stationary exactly when

$$P_{w_a}^{\perp} \left\{ \frac{1}{Z_a} \sum_b p_b g(w_a^\mathsf{T} S w_b) V w_b + M(w_a) \right\} = 0 \quad \text{for every } a. \tag{56}$$

Equivalently, the expression inside the projection can be multiplied by Z_a, in which case the MLP contribution is $Z_a M(w_a)$. Omitting this factor would change the balance between attention and the MLP. Integer token counts at each vertex therefore become real masses. Passing to the integral enlarges the family because mass ratios need no longer be rational.

11.2 Wasserstein geometry

Wasserstein distance measures the minimum cost of moving one mass cloud into another. It is naturally suited to (54). It does not automatically turn every attention system into energy descent. Under the tied condition $S = V = V^\mathsf{T}$, an antiderivative G of g gives the interaction energy

$$\mathcal{E}_G(\mu) = \frac{1}{2} \iint G(x^\mathsf{T} V y)\, d\mu(x) d\mu(y).$$

Unnormalized attention is its projected gradient. If $g > 0$, row normalization multiplies each token gradient by the positive mobility $1/Z_\mu(x)$, so the same energy remains monotone. Untied or antisymmetric matrices can remove every common energy.

12 Changing the exponential or normalization

For $V = \mathrm{diag}(2, -3)$ and three equal masses, the four tested kernels are

$$g_{\exp}(s) = \mathrm{e}^{1.5s}, \qquad g_{\mathrm{sig}}(s) = \frac{2}{1 + \mathrm{e}^{-1.5s}}, \qquad g_{\mathrm{sp}}(s) = \frac{\log(1 + \mathrm{e}^{1.5s})}{\log 2}, \qquad g_{\mathrm{poly}}(s) = \left(1 + \frac{s}{3.01}\right)^4.$$

The score interval is $[-3, 3]$, so each kernel is strictly positive there and is normalized to equal one at zero.

Kernel	mixed three-group roots q	stable root
exponential	$-0.907,\ 0.024,\ 0.801$	0.024
bounded sigmoid	-0.680	none
softplus	-0.787	none
polynomial $(1 + s/3.01)^4$	$-0.842,\ 0.231,\ 0.548$	0.231

Mixed three-group branches also exist for the other kernels, with different locations and stability. In the attention-only equation, removing the denominator for a positive kernel preserves the equilibrium equation token by token. It preserves full trajectories up to a single time reparametrization only when one token moves or when all row denominators agree along the orbit. For several moving tokens, the row factors can change paths and basins. In the presence of an MLP, several heads, or any other additive field, the denominator can also move equilibria by changing the relative strengths of the summed terms.

At a common attention-only equilibrium, the two Jacobians satisfy

$$J_{\text{unnorm}} = \operatorname{Diag}(Z_i I_{d-1}) J_{\text{norm}},$$

because the derivative of Z_i multiplies a vanishing tangent field. In the tied gradient case, positive mobility preserves the Morse index of a hyperbolic critical point. For untied S, V, left multiplication by unequal positive row factors need not preserve eigenvalue real parts, so local stability can also change.

Removing the sphere is much more radical. For a single token initialized at unit norm on the eigendirection of eigenvalue 2, spherical projection removes the whole push. Without that projection, normalized attention gives $r' = 2r$, so the norm reaches 10 at time $\log(10)/2 = 1.151\ldots$. Removing the attention denominator as well gives $r' = 2r\exp(3r^2)$ under the tested exponential parametrization. Its exact blow-up time is

$$\int_1^\infty \frac{dr}{2r\exp(3r^2)} = \frac{1}{4}E_1(3) = 0.003262\ldots.$$

A radial push that was invisible on the sphere then controls the norm dynamics.

13 State variables and observables

Attention weights alone do not describe the state. For each head, velocity depends on the two functions

$$Z_\ell^\mu(x) = \int k_\ell(x, y) d\mu(y), \qquad N_\ell^\mu(x) = \int k_\ell(x, y) C_\ell y \, d\mu(y). \tag{57}$$

Head ℓ uses N_ℓ / Z_ℓ. One must therefore observe both who is selected and what is transported. With a quadratic MLP, one must also retain the hidden measurements $u_r^\mathsf{T} x_i + c_r$ and the linear projections used at the output.

For the aligned symmetric case, $H = XVX^\mathsf{T}$ and its pieces within each eigenspace give a closed description up to rotations that commute with V. In the general anisotropic case, XX^T loses the orientation of the tokens relative to S, V, and the MLP weights. The following observables determine the instantaneous velocity; no minimality or autonomous-closure claim is made:

1. the full oriented token matrix X (a Gram matrix alone is insufficient);
2. the score matrices and weights of each head;

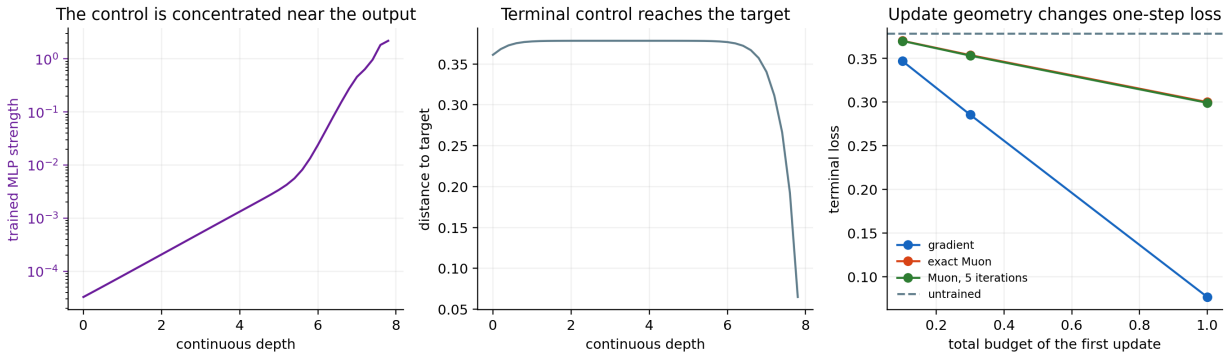


Figure 9: Left: norm of the trained MLP control along continuous depth. Center: distance to the rotated target. Right: terminal loss after one update at equal global norm for gradient descent, exact Muon, and five Newton–Schulz iterations.

3. directions after value and output transport;
4. hidden MLP features;
5. for a supervised task, positions, labels, and the readout.

For the exponential kernel with symmetric positive-semidefinite S, the scalar score kernel has the feature expansion

$$e^{\beta x^\top S y} = \sum_{q=0}^{\infty} \frac{\beta^q}{q!} \left\langle (S^{1/2}x)^{\otimes q}, (S^{1/2}y)^{\otimes q} \right\rangle . \tag{58}$$

Thus Z_ℓ^μ is a scalar kernel mean and N_ℓ^μ is its value-weighted, vector-valued analogue. The latter cannot be reconstructed from attention weights alone. If S is indefinite or nonsymmetric, a single positive-definite feature map need not exist; the paired query and key maps, together with the value attached to the same token, remain sufficient.

The hyperchaotic example makes this measurable: the maximum temporal standard deviation of the weights is only 0.0034, whereas that of the Gram geometry is 0.481. In the uniform-attention example, weight distance is exactly zero at every step even when geometric distance is not.

14 Training-time control and weight spectra

Isobe, Inoue, and Imaizumi fix unnormalized symmetric attention, add noise, and train only an FFN that is linear in its parameters under a quadratic cost [26]. Their optimal solution has three phases: tokens cluster, remain near a well for a long time, and then the FFN acts strongly near the output to reduce terminal loss. Their attention is parallel to the FFN in the continuous equation, whereas our MLP is serial in the finite layer. Section 2 explains why the two models agree to first order and differ layer by layer at finite step size.

14.1 Trained control around a polygon

We train a small angular control, linear in five Fourier features, to move a stable polygon toward a target rotated by 0.9 radians. Maximum control in the first half is 0.0011; in the last five intervals it reaches 2.19. Terminal error is 0.00133. The force is therefore about 2,000 times larger at the end than at the beginning. The same late-control profile occurs in this polygon experiment.

This result suggests a design principle: fix an attention dynamics that creates useful basins, then train a serial MLP to remain in a basin, leave it, or move its destination. It does not guarantee that every algebraic condition can be encoded by a small MLP or that training globally finds the correct weights.

14.2 Neural quadratic forms near initialization

The preceding sections evolve token states with fixed weights. The neural quadratic-form (NQF) theory of Liu, Xu, Poggio, and Chuang concerns the evolution of the weights during training [32]. The distinction matters because the two problems have different state spaces and different spectra.

Let $W = (w_1, \dots, w_r) \in \mathbb{R}^{p\times r}$ collect r exchangeable components of a scalar module evaluated on an input ξ. Under three-times differentiability, permutation symmetry of the components, and zero component gradient when that component is zero, their theorem gives

$$f_\xi(W) = f_\xi(0) + \operatorname{Tr}(\mathcal{M}_W A(\xi)) + O(\|W\|^3), \qquad \mathcal{M}_W = WW^{\mathsf{T}} \succeq 0, \tag{59}$$

where the symmetric structure matrix $A(\xi) \in \mathbb{R}^{p\times p}$ is determined by the architecture and the input. This statement applies to a factorized parameterization whose exchangeable components satisfy the hypotheses. Training the effective matrices S and V directly introduces linear parameter sectors and is a different model. Residual sphere normalization is smooth around a unit base token: $\mathcal{N}(x+\delta) = x + P_x^{\perp}\delta + O(\|\delta\|^2)$. It therefore projects the leading structure matrix as long as the pre-normalization vector stays nonzero; a nonresidual normalization at the origin would require a separate analysis.

For attention, the order at which a parameter appears depends on what is trained. Let $X_c \in \mathbb{R}^{d\times n}$ be a context matrix, with $\bar{x} = X_c\mathbf{1}/n$ and $\Sigma_{X_c} = X_cX_c^{\mathsf{T}}/n - \bar{x}\bar{x}^{\mathsf{T}}$. Put $Q = \varepsilon\widehat{Q}$, $K = \varepsilon\widehat{K}$, $V = \varepsilon\widehat{V}$, and $o = \varepsilon\widehat{o}$. Direct softmax expansion at inverse temperature β gives

$$f_{\text{full}}(\varepsilon) = \varepsilon^2\widehat{o}^{\mathsf{T}}\widehat{V}\bar{x} + \frac{\beta\varepsilon^4}{\sqrt{d_k}}\widehat{o}^{\mathsf{T}}\widehat{V}\Sigma_{X_c}\widehat{K}^{\mathsf{T}}\widehat{Q}x + O(\varepsilon^6). \tag{60}$$

Thus value–readout learning is quadratic, while query–key routing first enters at quartic order. The quadratic full-head structure matrix has a bipartite value–readout block, hence paired positive and negative eigenvalues, together with a large zero sector containing the query and key directions. If the value and readout are fixed and merged into c, training only Q and K instead gives

$$f_{QK}(\varepsilon) = c^{\mathsf{T}}\bar{x} + \frac{\beta\varepsilon^2}{\sqrt{d_k}}(\Sigma_{X_c}c)^{\mathsf{T}}\widehat{K}^{\mathsf{T}}\widehat{Q}x + O(\varepsilon^4). \tag{61}$$

Under column-wise vectorization, the corresponding structure matrix has the off-diagonal block

$$\frac{\beta}{2\sqrt{d_k}}\left[x(\Sigma_{X_c}c)^{\mathsf{T}}\right] \otimes I_{d_k},$$

with its transpose in the opposite block. A different vectorization permutes rows and columns. The spectral experiment below sets $\beta = 1$.

For mean-squared loss on samples (ξ_μ, y_μ), the pure quadratic model has

$$\mathcal{L}(W) = \frac{1}{m}\sum_{\mu=1}^{m}\left[\operatorname{Tr}(\mathcal{M}_W A_\mu) - y_\mu\right]^2, \tag{62}$$

$$\dot{W} = -\mathcal{H}_W W, \qquad \dot{\mathcal{M}}_W = -\mathcal{H}_W\mathcal{M}_W - \mathcal{M}_W\mathcal{H}_W, \tag{63}$$

$$\mathcal{H}_W = \frac{4}{m}\sum_{\mu=1}^{m}\left[\operatorname{Tr}(\mathcal{M}_W A_\mu) - y_\mu\right]A_\mu. \tag{64}$$

If the A_μ commute, write $A_\mu = P\operatorname{Diag}(\lambda_{\mu,1}, \dots, \lambda_{\mu,p})P^{\mathsf{T}}$ and $z_k = (P^{\mathsf{T}}\mathcal{M}_W P)_{kk}$. Then

$$\dot{z}_k = \frac{8}{m}\left[\sum_\mu y_\mu\lambda_{\mu,k} - \sum_j\left(\sum_\mu \lambda_{\mu,k}\lambda_{\mu,j}\right)z_j\right]z_k. \tag{65}$$

This is the generalized Lotka–Volterra reduction proved in the NQF paper.

Equation (65) organizes every sign pattern at initialization. With

$$r_k = \frac{8}{m} \sum_{\mu} y_\mu \lambda_{\mu,k}, \tag{66}$$

a mode with $r_k > 0$ initially grows, a mode with $r_k < 0$ initially decays, and a zero-rate mode is decided by nonlinear coupling. In a generic coupled Lotka–Volterra system, later competition can suppress or delay a mode despite its positive initial rate.

The logarithmic ignition formula requires one of the solvable reductions in the NQF analysis: orthogonal features, or isotropic samples with $\mathcal{M}_W(0)$ and the target operator simultaneously diagonalizable. Let $\zeta_k > 0$ denote the positive target or generator coefficient in the convention of the corresponding reduction; the linear growth exponent of z_k is $a\zeta_k$, with $a = 1$ for orthogonal features and $a = 2$ for isotropic samples. If $z_k(0) = \epsilon\rho_k$ with ρ_k bounded above and below independently of ϵ, then

$$t_k^\star = \frac{1}{a\zeta_k} \log \frac{1}{\epsilon} + O(1), \qquad a = \begin{cases} 1, & \text{orthogonal features,} \\ 2, & \text{isotropic samples.} \end{cases} \tag{67}$$

Equal ζ_k have no separation proportional to $\log(1/\epsilon)$; distinct positive rates do. Here ϵ is the initial scale of an eigenvalue or diagonal mode of $\mathcal{M}_W$. If entries of W instead have standard deviation σ, then typically $\epsilon \propto r\sigma^2$, so the slope against $\log(1/\sigma)$ is twice the slope in (67). Gaussian and norm-fixed initializations change the $O(1)$ intercept through ρ_k; they should leave the logarithmic slope unchanged in the quadratic gradient-flow regime. A normalization layer can change A_μ or violate the smoothness assumptions, and Muon's polar update is not governed by (63).

There is a conditional link to inference-space bifurcations. Suppose a scalar mode z_k in one of the solvable reductions parametrizes a frozen-weight equilibrium branch, and a simple eigenvalue $\ell(z_k)$ of its token-state Jacobian satisfies $\ell(z_c) = 0$ and $\ell'(z_c) \neq 0$ at a fixed threshold below the mode's plateau. The training time at which that branch changes stability then obeys

$$t_{\text{bif}} = \frac{1}{a\zeta_k} \log \frac{1}{\epsilon} + O(1). \tag{68}$$

Two bifurcations controlled by distinct growth rates therefore acquire a separation of order $\log(1/\epsilon)$; an exact rate degeneracy removes that leading separation. This implication requires an identified map from the NQF mode to the state Jacobian. Neither the NQF theorem nor the training experiment of Lai et al. supplies that map for a general Transformer.

The saddle–node in (48) gives a second conditional case. If a learned MLP multiplier satisfies $\Gamma(t) = \Gamma_0 + b_\Gamma z_k(t)$ with $b_\Gamma > 0$, and the threshold $z_c = (\gamma_c - \Gamma_0)/b_\Gamma$ is fixed independently of ϵ and satisfies $0 < z_c < z_{k,\infty}$, then the time at which the synchronized pair is created still has the logarithmic form $(a\zeta_k)^{-1} \log(1/\epsilon) + O(1)$. The post-creation root separation and Jacobian eigenvalue magnitude are both $\Theta(\sqrt{\Gamma - \gamma_c})$; this is not the simple linear eigenvalue crossing assumed in the preceding paragraph. This is an implication of the assumed scalar coupling, not a claim that generic Transformer training follows that path.

The NQF scaling theorem additionally takes an infinite-mode limit and assumes $\alpha_1 > 1$, $\alpha_2 > 0$, and controlled initial coefficients. In the theorem, the effective growth rates and target strengths have the exact tails $\zeta_k = c_\zeta k^{-\alpha_2}$ and $\mathcal{V}_k = c_\mathcal{V} k^{-\alpha_1}$. If $z_k(0) = \epsilon c_k k^{-b}$ with c_k bounded above and below and $b \geq \alpha_1 - \alpha_2$, then, first taking the number of modes to infinity at fixed ϵ and then taking $\epsilon \to 0$, the excess loss at $\tau = t/\log(1/\epsilon)$ scales as

$$\mathcal{E}(\tau) = \Theta\left(\tau^{-(\alpha_1 - 1)/\alpha_2}\right). \tag{69}$$

for almost every rescaled time τ in the theorem's domain. The architecture determines the structure matrices; it does not imply these power-law tails.

14.3 Numerical spectral atlas

We tested the preceding statements in the exactly solvable isotropic NQF, rather than in a generic trained Transformer where the quadratic approximation would be local. For every dimension $p \le 6$, the experiment enumerates every inertia and every partition of the positive and negative eigenvalue multiplicities. There are 280 such categorical classes. Continuous eigenvalue magnitudes are not enumerated. Write γ_k for the eigenvalues of the symmetric label-weighted generator $Y = 4m^{-1}\sum_\mu y_\mu A_\mu$. Across all classes, the maximum error between the fitted initial rate of $\log z_k$ and the predicted value $2\gamma_k$ is 1.23×10^{-7}.

Ignition was measured for generator eigenvalues $(1, 0.7, 0.7, 0.25)$, widths 32, scales $\epsilon = 10^{-2}, \dots, 10^{-8}$, and 64 paired replicates under Gaussian, row-norm- fixed, and row-Stiefel initializations. The largest relative error in the fitted slopes of $t_k^\star$ against $\log(1/\epsilon)$ is 0.101%. The two modes with eigenvalue 0.7 have a fitted difference slope of magnitude at most 1.60×10^{-4}, consistent with the absence of logarithmic separation at an exact degeneracy. Initialization changes the intercept and its variance. It does not change the slope within this solvable regime. Individual eigenvectors inside a degenerate eigenspace are not identifiable, so their comparison uses projector overlap; outside degeneracies, the leading eigendirections of $\mathcal{M}_W$ align with those of the generator during growth.

The attention expansion gives a separate structural check. For a random five-dimensional context of length 11, key dimension 3, value dimension 4, and 48 samples, the full-head quadratic generator has inertia $(4, 46, 4)$: four positive, 46 zero, and four negative eigenvalues. Removing softmax normalization preserves that inertia and multiplies its spectral radius by 11 in this zeroth-order value–readout sector. With fixed values and readout, the QK generator has inertia $(15, 0, 15)$; positive and negative eigenvalues pair to within 2.9×10^{-15}. These are structure-matrix spectra at initialization, not the eigenvalues of V or a token-state Jacobian.

For the NQF tail assumptions, 16,384 modes with $(\alpha_1, \alpha_2) = (2, 0.8)$ give a fitted excess-loss exponent -1.25623, compared with the prediction -1.25. Exponential-rate and flat-rate controls do not exhibit the same stable power-law slope. Finally, a spiked-Wishart control with aspect $\gamma_{\text{asp}} = p/r = 1/2$ separates the BBP claim from the NQF dynamics. Its population spike satisfies $1 + \theta(t) = \exp(2\delta t)$ with $\delta = 0.25$, and hence $t_c = \log(1 + \sqrt{\gamma_{\text{asp}}})/(2\delta)$. At $p = 256$, $r = 512$, and 32 replicates, the null top eigenvalue is 2.8678 versus the asymptotic bulk edge 2.9142, and its mean squared spike overlap is 0.00187. At twice the critical time, the measured outlier is 3.6453 versus 3.6754, and the overlap is 0.6861 versus 0.6847. A row-Stiefel initialization has $\mathcal{M}_W(0) = \epsilon I$ and no finite-width Wishart bulk; its planted direction is selected for every $t > 0$ after anisotropic growth begins, although no direction is distinguished at $t = 0$. The BBP transition is therefore initialization-ensemble specific, whereas the logarithmic NQF clocks persist across the three tested initializations.

14.4 Three spectra that should not be identified

The eigenvalues of V in theorem 3.2 classify stationary token geometry in the tied fixed-weight flow. Eigenvalues of the tangent Jacobian and Floquet or Lyapunov exponents determine inference-depth stability. Eigenvalues of A_μ, $\mathcal{H}_W$, and $\mathcal{M}_W$ determine training-time parameter growth in the NQF approximation. No general identity relates these spectra. An empirical comparison must measure their eigenvector overlaps as well as their eigenvalues.

A Baik–Ben Arous–Péché (BBP) outlier requires a further random-matrix model [8, 37]. For the standard rank-one spiked covariance control with aspect ratio $\gamma_{\text{asp}} = p/r$ and population covariance $I + \theta uu^\mathsf{T}$, the top sample eigenvalue separates from the Marchenko–Pastur bulk only for $\theta > \sqrt{\gamma_{\text{asp}}}$. The limiting bulk edge is $(1 + \sqrt{\gamma_{\text{asp}}})^2$, while the outlier limit is $(1 + \theta)(1 + \gamma_{\text{asp}}/\theta)$. The low-dimensional V spectra used for the attractor examples have no bulk limit and admit no

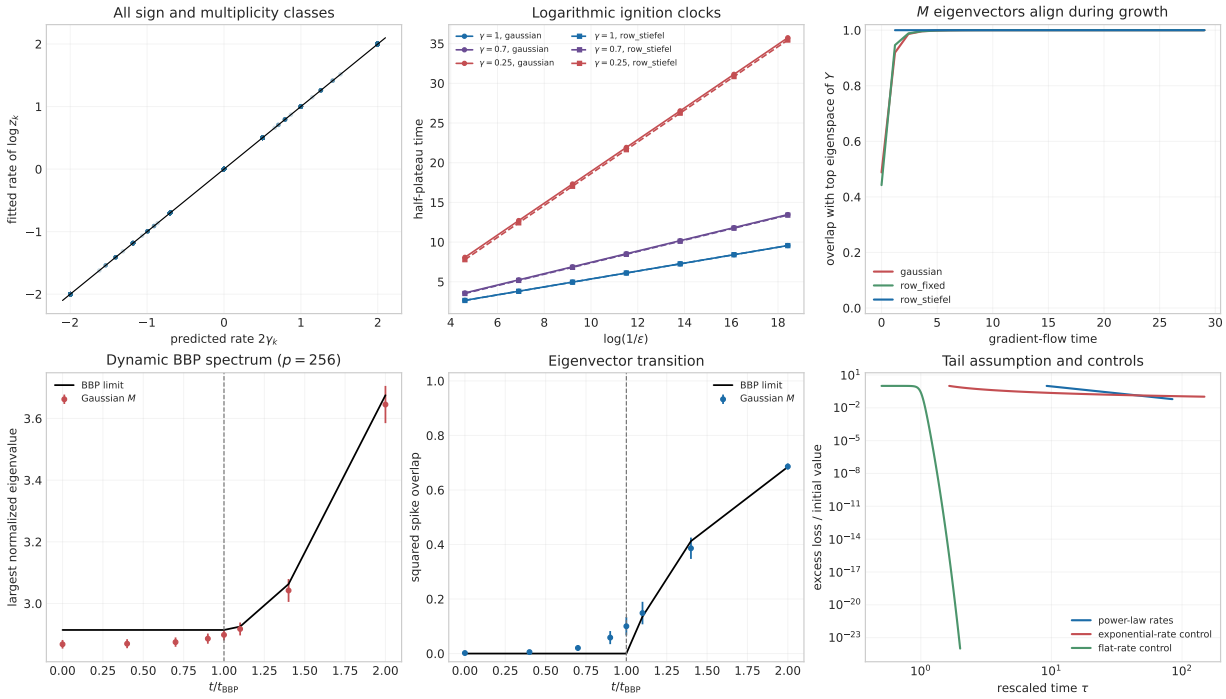


Figure 10: Spectral audit of the solvable isotropic NQF. Top row: predicted and fitted initial mode rates over all 280 categorical spectra; ignition clocks for three initializations; alignment of identifiable eigenspaces of $\mathcal{M}_W$ with the generator. Bottom row: top eigenvalue and spike overlap in the spiked-Wishart control, followed by the power-law tail and two non-power-law controls. Error bars are standard errors over 32 Gaussian matrices in the $p = 256$ BBP panels.

BBP interpretation.

14.5 A first Muon step

If the gradient of a matrix at depth k is $G_k = U_k \Sigma_k R_k^\top$, a gradient step is $-\alpha G_k$. An idealized Muon step instead uses

$$-\eta U_k R_k^\top. \tag{70}$$

It preserves singular directions and equalizes nonzero magnitudes. At the first step from zero momentum, momentum only changes the global gradient scale, which this polar factor then removes. Under an operator-norm constraint per matrix, this is the steepest direction; under the L^2 constraint of Imaizumi's theorem, the gradient is naturally favored.

In our 40-slice analogue, the final gradient is 7.11×10^{11} times larger than the initial gradient. At total budget 0.3, loss falls from 0.37839 without training to 0.28547 with gradient descent, 0.35385 with exact Muon, and 0.35323 with five Newton–Schulz iterations. At budget 1.0, Muon acts much earlier and destroys the plateau, while the gradient still concentrates its action on the last slice. This is not a universal ranking: it shows that the norm used to compare updates determines the winner.

Recent associative-memory analyses give another explanation for Muon's usefulness: it balances learning of rare directions more evenly than anisotropic updates [46, 28]. Our experiment does not replace those results; it shows their analogue along the depth of a polygonal dynamics.

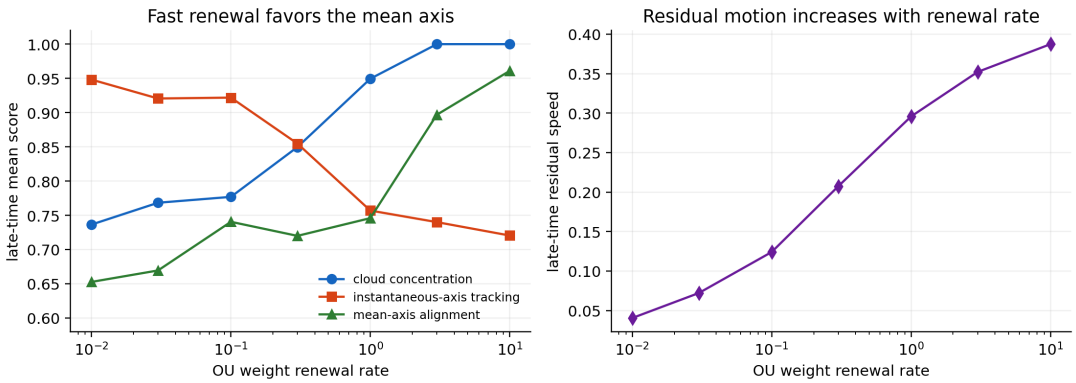


Figure 11: Late-time statistics versus the OU renewal rate for 32 tokens, four heads, 12 repetitions, $\sigma = 0.65$, and integration step 0.02. Left: concentration and alignment statistics. Right: residual token speed.

15 Slow Ornstein–Uhlenbeck weight dynamics

Massucco and collaborators represent heads by a distribution of matrices that changes with depth [34]. In their Ornstein–Uhlenbeck regime,

$$d\Theta_t = \kappa(F - \Theta_t)dt + \sqrt{2\kappa}\,\sigma\,dB_t, \qquad \kappa > 0, \tag{71}$$

the weight law returns at rate κ to a Gaussian centered at F with entrywise stationary variance σ^2. The simulator uses the exact transition

$$\Theta_{t+\Delta t} = F + \mathrm{e}^{-\kappa\Delta t}(\Theta_t - F) + \sigma\sqrt{1 - \mathrm{e}^{-2\kappa\Delta t}}\,\Xi_t.$$

At $\kappa = 0$, it draws $\Theta_0 = F + \sigma\Xi_0$ and freezes that draw. Massucco and collaborators prove asymptotic stationarity under compact parameter support and a bound on accumulated explicit time variation of the energy. A Gaussian OU path has unbounded support and nonintegrable path variation, so that theorem does not establish relaxation for the literal stochastic process above. Their OU discussion gives a formal finite-head/noise estimate and numerical evidence; it does not cover the slow sample-path regime studied here.

Our first approach isolates renewal speed. It uses 32 tokens on $\mathbb{S}^2$, four heads, 12 repetitions, $\sigma = 0.65$, step 0.02, and 2,000 steps. The target mean matrix has eigenvalues $(1.2, 1.0, 0.2)$. The OU rate ranges from 0 to 10.

At rate 0.01, late concentration is 0.736 and instantaneous-axis tracking is 0.948. At rate 0.3, they are 0.850 and 0.855. At rate 3, concentration is nearly one, but instantaneous tracking is only 0.740, while mean-axis alignment is 0.897. Late speed grows from 0.040 to 0.352. This exposes three time scales: token contraction, landscape motion, and noise averaging.

If the frozen system has a uniformly attracting equilibrium branch, with a relaxation gap bounded away from zero and no nearby bifurcation, sufficiently slow weight motion gives quasistatic tracking. When relaxation and renewal times become comparable, the numerical sweep exhibits lag and path dependence; basin changes can occur near a stability boundary. The sweep measures lag and concentration. It does not establish a chaotic attractor caused by slow OU weights.

The saddle–node in (49) quantifies the loss of this separation. With $\mu = \gamma - \gamma_c > 0$, the stable-branch gap is $\Theta(\sqrt{\mu})$ and $|d\theta_\star/d\gamma| = \Theta(\mu^{-1/2})$. For a smooth deterministic parameter path, linear-response power counting gives a tracking lag of order $|\dot\gamma|/\mu$. Keeping this lag small relative to the branch separation $\Theta(\sqrt{\mu})$ requires

$$|\dot\gamma| \ll \mu^{3/2}. \tag{72}$$

This condition necessarily fails in a shrinking neighborhood of the bifurcation for any fixed nonzero sweep rate. Brownian OU paths are not differentiable, so (72) is not a samplewise criterion for (71). Applying the scalar condition to matrix-valued OU dynamics would also require identifying the coordinate of Θ_t that controls γ; the corresponding noisy first-passage problem remains open here.

16 Relations to statistical-physics models

For $\beta > 0$, each softmax row is exactly a conditional Boltzmann law:

$$a_{ij} = \frac{\mathrm{e}^{-E_{ij}/T}}{\sum_k \mathrm{e}^{-E_{ik}/T}}, \qquad E_{ij} = -x_i^{\mathsf{T}} S x_j, \qquad T = 1/\beta.$$

Uniform attention at $\beta = 0$ is the infinite-temperature limit. This alone does not make the system equilibrium matter: the weights are Boltzmann probabilities, but the token trajectory is deterministic. A genuine heat bath would require tangent Brownian noise. If the field is a gradient and noise is matched to its mobility, the stationary law is Gibbs. With antisymmetric transport or a general MLP, currents can persist and the system remains out of equilibrium.

The tied isotropic choice $S = V = I$ is a mean-field $O(d)$ vector-spin model on the sphere with an exponential ferromagnetic pair interaction, followed by deterministic zero-noise gradient ascent with token-dependent mobility. Several tied symmetric heads give a sum of anisotropic vector-spin interactions only when their mobilities admit a common energy. Fixed random symmetric heads are quenched anisotropy rather than independent Sherrington–Kirkpatrick couplings. Untied S and V give a nonreciprocal nonequilibrium system in general.

The appearance of tanh in the bipolar criteria has the elementary two-state Boltzmann origin

$$\frac{\mathrm{e}^{a}}{\mathrm{e}^{a} + \mathrm{e}^{-a}} = \frac{1 + \tanh a}{2}.$$

It does not by itself imply a thermodynamic phase transition.

Modern Hopfield attention evolves one query against fixed stored patterns and decreases a log-sum-exp energy [39]. Here every token moves, the interaction is normalized separately by row, and value transport can differ from the score geometry. On the circle there is an exact oscillator representation. If $t_i = Jx_i$ and $V = aI + bJ$, then

$$t_i^{\mathsf{T}} V x_j = a \sin(\theta_j - \theta_i) + b \cos(\theta_j - \theta_i) = \sqrt{a^2 + b^2} \sin(\theta_j - \theta_i + \varphi), \quad \varphi = \operatorname{atan2}(b, a). \tag{73}$$

The antisymmetric value component therefore has the Sakaguchi–Kuramoto phase-lag form [42], although the softmax makes its pair weights state-dependent. A quadratic MLP supplies external harmonics up to order three by proposition 2.1. Nunley's frustrated-synchronization architecture uses this oscillator structure directly, with trained torus phases, multiple harmonics, and a causal delay [36]; it is architecturally different from the spherical flow studied here. Higher-dimensional reciprocal and rotational sectors are related to Heisenberg spins and Lohe oscillators, respectively [33], but none of these models equals the general serial block.

A replica analysis would require a precise random ensemble of matrices, a joint limit of token count, dimension, and number of heads, and an overlap observable. The Gram matrix between two replicas subjected to the same weights is a natural candidate, but attention weights are insufficient: values and MLP features must also enter. The works cited here do not provide a replica solution for our exact serial block. Rende, Gerace, Laio, and Goldt map a simplified attention learner to a generalized Potts model and compute generalization under a replica-symmetric ansatz [40]; their calculation relaxes the Potts variables to Gaussian magnetizations and studies convex ridge square loss. Tiberi, Mignacco, Irie, and Sompolinsky derive a Bayesian thermodynamic theory for

simplified fixed-query/key attention paths [43]. Replica-symmetry breaking has been worked out for dense associative memories [3], but transferring it here would require a quenched ensemble, a Gibbs measure, a thermodynamic limit, and a nontrivial distribution of replica overlaps. The conditional Boltzmann form of one softmax row supplies none of these by itself.

17 Relation to Altafini and the tied symmetric model

Altafini studies attention dynamics and its multistability without a serial MLP and explicitly introduces Q, K, and V [6]. For frozen dynamics, Q and K enter through the product $Q^\mathsf{T}K$, as they do here. His main spectral assumption is that the symmetric value matrix has a simple largest eigenvalue $\lambda_1 > 0$; it neither requires $Q^\mathsf{T}K = V$ nor makes the score matrix symmetric. His polygonal class has zero received output. The instability theorem for that class does not cover the nonzero radial-output equilibria above or a serial MLP. Altafini also reports that a nonsymmetric V with a leading complex-conjugate eigenpair can approach a stable cycle. Thus the attention-only cycle in section 7 is not the first observation of oscillation in an untied model. Its additional content is an explicit reproducible matrix instance, changing Gram geometry, long-horizon tangent-spectrum estimates, perturbation returns, and finite-layer period scaling.

Kuehn and Yoon impose $Q^\mathsf{T}K = V = V^\mathsf{T}$ and obtain eigenspace selection and a gradient flow structure [29]. Polyanskiy, Rigollet, and Yao prove that the homogeneous reciprocal circle model synchronizes from almost every initial state for $\beta \geq -0.16$, hence throughout the positive-temperature range used here [38]. Stable nonconsensus examples in this paper use anisotropic V, untied transport, or an MLP and lie outside that theorem.

The serial-block analysis proves or numerically tests:

1. the exact condition for all fixed points and cycles of the serial attention–MLP block;
2. an experimental taxonomy of the four reciprocal/untied and potential/general combinations;
3. a proof that every accumulation point of a fixed-parameter, fixed-period family is stationary as $h \to 0$, followed by an audit down to $h/4096$;
4. two exact synchronized cycles, numerical internal-cycle candidates, two three-token chaotic-attractor candidates, and one eight-token hyperchaotic-attractor candidate in the complete block;
5. a demonstration that constant attention weights can hide internal motion;
6. multihead maps in which the sum creates new destinations;
7. an experimental separation of orthogonality, uniformization, and genuine decoupling in high dimension;
8. the passage from finite polygons to atomic measures, kernel changes, trained control, one Muon step, and the first slow-OU sweep.

The equilibrium equation, $1/h$ period scaling, and tangent spectrum separate fixed destinations, finite-layer cycles that collapse as $h \to 0$, and recurrent solutions of the continuous-depth equation.

18 Stationary equations as constrained computational problems

The exact solutions produced by the dynamics are unit vectors satisfying (34), or periodic sequences satisfying (32). Choosing S, V, and M therefore chooses a family of nonlinear equations on a product of spheres. A Euclidean projected-gradient objective for the total field is used in three cases here: the MLP alone when $M = \nabla\phi$; uniform tied attention ($\beta = 0$) plus such an MLP; and unnormalized tied attention plus such an MLP. Row-normalized tied attention alone has the Lyapunov energy (17) with token-dependent positive mobilities. Adding an unscaled potential MLP at $\beta > 0$ generally destroys the common gradient structure even though each subfield separately

comes from a potential.

Several standard hard equations occur as exact special cases or close algebraic relatives. Turn attention off and take one token with $b = L = 0$ and $c_r = 0$. For a homogeneous potential quadratic MLP $M(x) = \mathcal{T}(x, x, \cdot)$, (34) reads

$$\mathcal{T}(x, x, \cdot) = \lambda x, \qquad \|x\| = 1, \tag{74}$$

for a symmetric order-three tensor $\mathcal{T}$. This is the tensor Z-eigenvector equation. Computing extrema and spectral quantities for order-three tensors is NP-hard in general [24]. The projected flow searches for a stationary tensor direction; it gives no guarantee of finding the global extremum.

The low-rank semidefinite factorization

$$\min_{Y} \langle C, YY^{\mathsf{T}} \rangle \quad \text{subject to} \quad \operatorname{diag}(YY^{\mathsf{T}}) = \mathbf{1} \tag{75}$$

may be taken with $C = C^{\mathsf{T}}$ and has the KKT equation $CY = \operatorname{Diag}(\lambda)Y$ [11]. For scalar $V = \nu I$, our tied equilibrium equation has the same row-eigenvector form, $\nu R(X)X = \Lambda_X X$, but with $R(X) = \exp_\circ(\beta\nu XX^{\mathsf{T}})$. The entries of $R(X)$ are positive, so for $\nu \neq 0$ all entries of $\nu R(X)$ have one sign; the matrix also depends on the unknown X and need not be positive semidefinite. It cannot represent an arbitrary signed matrix C. Complex phase synchronization instead uses $C = C^*$ and imposes $(Cz)_i = \lambda_i z_i$ with $|z_i| = 1$ and real multipliers λ_i [10]. These correspondences identify the equations solved at a fixed point; they do not transfer global convergence or complexity guarantees.

Alcalde, Ji, and Zuazua obtain a stronger control result in a different model [5]. For Euclidean attention with an affine feed-forward field, Gaussian input laws remain Gaussian and the dynamics closes on their mean and covariance. Time-dependent controls reach exactly the Gaussian targets with the same covariance rank. Their result concerns finite-dimensional moment equations without sphere normalization; arbitrary finite token configurations are outside its scope.

Associative-memory retrieval, spherical codes, phase synchronization, and graph objectives can therefore be encoded only after specifying the corresponding potential or inverse design of S, V, M. In particular, embedding the MaxCut energy in a product-of-spheres objective leaves a nonconvex search problem. Convergence to a stationary point, or escape from one basin under OU noise, does not imply a maximum cut. Nothing proved or measured here yields a polynomial-time algorithm for an NP-hard problem.

19 Open analytical and empirical questions

The unresolved questions with direct bearing on the present conclusions are:

1. reproduce cycles and chaos with realistic head dimensions, explicit W^O matrices, and RMSNorm rather than an ideal sphere;
2. train the full serial block instead of a reduced FFN control on a task with out-of-sample validation;
3. vary token count, dimension, temperature, and head count jointly to separate mean-field, head, and dimensional limits;
4. map bifurcations when OU weights evolve as slowly as token relaxation, including hysteresis and basin-jump measurements;
5. test whether the numerical chaotic-attractor candidates persist after adding noise, bias, affine LayerNorm, and reduced precision;
6. refine a fixed initial-state window through the saddle–node crossover and test terminal-basin labels or recurrent folding separately from thresholded settling times;
7. construct a two-replica experiment under shared weights and measure the full overlap including values and MLP features;

8. prove an approximation bound relating MLP width and degree to realizable tangent fields on the sphere;
9. separate optimization performance from total cost for Muon, gradient descent, and Adam on the same depth-control problem.

20 Conclusion

Equation (34) characterizes the stationary states of the serial attention–MLP block in the continuous-depth limit. Finite-layer periodic orbits satisfy (32); a fixed discrete period collapses to a stationary state as $h \to 0$. The observed internal cycles instead have a layer period proportional to h^{-1}, and their return times approach those measured by direct integration of (33). This, together with transverse spectra and perturbation returns, is numerical evidence for continuous periodic orbits. The same tangent tests assess attraction, while variation of the Gram geometry distinguishes internal cycles from rigid rotations. Positive tangent rates provide evidence for the two chaotic and one hyperchaotic attractor candidates reported here.

The reciprocal, potential family was classified as converged on the stated windows in all tested continuous-flow instances. Untied value transport or a general MLP permits persistent circulation. For multiple heads, this circulation and the equilibrium condition depend on the sum of head outputs. In high dimension, orthogonality coexists with nearly uniform attention at fixed β; in the many-token limit, the dynamics becomes (54). The trained-control, Muon, and OU experiments concern specific finite protocols and do not alter these analytical statements.

The settling-time atlas gives a negative result about the proposed fractal analogy. Its fitted slopes decrease under co-moving zoom, but vary with threshold, grid, and horizon; the set excluded by the combined audit is one nearly affine strip, and the full-interval tangent exponents for the three audited trajectories are negative. This behavior is quantitatively consistent with the synchronized saddle–node: a smooth hyperbolic saddle can have a saturated low finite-grid slope, and a smooth critical fold has asymptotic settling-time exponent one half. Nearby initial states can therefore require very different depths without a fractal basin boundary. A fractal claim for this model would require fixed-window refinement beyond the fold crossover together with terminal-basin or recurrent-folding evidence.

A Numerical ensembles and decision rules

A.1 Finite-layer census

The broad census uses planar tokens. Independently for each model, the attention and MLP scales are log-uniform on $[0.08, 3]$. With probability 0.25, the common residual strength h is log-uniform on $[0.02, 0.30]$; otherwise it is uniform on $[0.30, 1.60]$. Sharpness is log-uniform on $[0.08, 12]$ and is then replaced by zero with probability 0.08. The five post-hoc bins have cut points

$$h : (0.10, 0.30, 0.60, 1.00), \qquad \beta : (0.10, 0.50, 2, 6), \qquad a_{\mathrm{attn}}/a_{\mathrm{MLP}} : (0.25, 0.75, 1.50, 4).$$

Raw score and value matrices have independent standard Gaussian entries and are divided by their entrywise root-mean-square. A tied draw symmetrizes and renormalizes V, sets $S = V$, and multiplies both by the attention scale. An untied draw chooses the four symmetry subtypes with equal probability, symmetrizes and renormalizes the applicable matrices, and applies the same attention scale to S and V.

The width is uniform on $\{1, 2, 4, 8\}$. Active hidden directions are normalized Gaussian vectors; their biases have law $N(0, 0.4^2)$. The MLP bias has law $N(0, 0.35^2 I)$. Linear coefficients are Gaussian

and are symmetrized in a potential MLP. Potential output columns are $\gamma_r u_r$ with $\gamma_r \sim N(0, 1/m)$; general output entries have law $N(0, 1/m)$. The MLP scale multiplies its bias, linear map, and output columns. Initial angles are independent uniform variables on $[-\pi, \pi)$.

Families 1, 2, and 4 use 1,024 models at each $n = 1, 2, 3, 4$; family 3 uses 4,096. The stored seeds are 260813002–260813005 and the batch size is 32. Each model has 96 starts. The implementation applies 1,400 full serial layers, comprising 2,800 normalized attention and MLP substeps, and retains the next 36 states. Candidate periods $p = 1, \dots, 12$ are tested in increasing order. The first p for which every overlapping pair of retained states has maximum wrapped coordinate error below 2×10^{-6} is recorded. If no period passes, a trajectory is called a rigid rotation when the maximum one-step Gram change and the spread of token increments are both below 2×10^{-5} and the common mean increment exceeds 2×10^{-5} in magnitude. All remaining trajectories are unresolved.

For each family and $p \in \{3, 4\}$, the first stored screen hit was refined by solving $F^p(\theta) - \theta = 0$ with nonlinear least squares at tolerances 10^{-13} and at most 4,000 function evaluations. The derivative of F^p uses centered differences of size 10^{-6}. A representative is primitive when its p-return residual is below 10^{-9} and every earlier residual exceeds 10^{-5}; it is declared stable when the spectral radius is below $1 - 10^{-6}$.

The fixed-point table uses the same model law, with 128 models in families 1, 2, and 4 and 256 in family 3. For $n = 1$ the solver starts from 60 equally spaced and 180 random angles; for $n = 2$ it uses a 10×10 grid and 180 random starts; for $n = 3$ it uses 180 random starts. Roots have wrapped residual at most 2×10^{-8} and are deduplicated after sorting tokens and rounding angles to five decimals. Consequently, the reported multistable counts for $n \geq 2$ are search-dependent lower bounds.

A.2 Weak-layer continuation and direct replay

The harvest uses the same parameter law, 1,500 burn layers, a 48-state recurrence tail, and 16 starts per model for $n \leq 4$. The numbers of configurations in families 1–4 are respectively 32,768, 16,384, 65,536, and 16,384. The $n = 8, 16$ supplement uses 24,576 configurations and eight starts per model. Two forced-$\beta = 0$ controls contribute 16,384 configurations each; they reuse the corresponding random streams before β is overwritten and are therefore paired controls. The 26,112 fresh finite-horizon models use one start each. These components give the totals in section 6.

The harvest retains at most the first qualifying basin of a model for each of the labels p3, p4, and chaotic candidate. The last label requires failure of the period and rigid-rotation screens and a one-vector Benettin rate greater than 0.01 over 500 layers, using a perturbation of size 10^{-7}. Reservoir sampling is followed by a shuffle and round-robin selection over token count and untied subtype, capped at 256 records per family and label. This gives family totals $(286, 378, 768, 476)$ and 1,908 records overall.

Continuation carries the terminal state from one ratio to the next over

$$1, .7, .5, .35, .25, .18, .125, .088, 1/16, .044, 1/32, .022, 1/64, .011, 1/128, .0055, \\ 1/256, 1/512, 1/1024, 1/2048, 1/4096.$$

At each ratio it adds Gaussian angular noise of standard deviation 10^{-7}, burns for $\lceil 600/r \rceil$ layers, and retains 128 states with stride $\lceil 0.25/r \rceil$, where r is the displayed ratio. A record is fixed when both motion and field RMS are below 10^{-3}; it is moving when motion is at least 10^{-3}, and recurrent when in addition the best 90%-quantile wrapped return error over saved lags 3–80 is below 0.03. The positive-rate threshold is 0.005.

The table in section 6 uses RK4 step 0.02, an initial angular perturbation of standard deviation 10^{-3}, 600 burn units, and two further perturb-and-relax stages of 200 units each. It retains 160

states at spacing 0.25; Lyapunov durations are 300 for p3/p4 seeds and 400 for chaotic-candidate seeds. These records are selected from the finite-layer screen and the displayed fractions remain conditional on that selection.

A.3 Attention-only cycle

Full-precision versions of the displayed matrices in (41) and the source trajectory are stored in `data/spectral_self_attention/non_mlp_report_results.json` and `data/spectral_self_attention/continuous_ode_f4_p3_cycle_attention_only.json`. A reproducible phase point is

$$(-0.993587062149, 1.097020681051, 2.830111454428).$$

The orbit is formed after a 120-unit RK4 burn at step 0.005 and sampled at step 0.01. The perturbation experiment uses seed 260816101 and 128 independent Gaussian angular kicks of standard deviation 0.08 about that phase. After 36 units at RK4 step 0.01, distance to the nearest of 912 stored orbit phases is at most 0.003673 radians, with median 0.001936; the corresponding initial maximum is 0.142421. This finite-window, phase-minimized return is the basis for the perturbation statement in the main text. The spectrum uses a separate 2,000-unit Benettin–QR run at step 0.01, tangent increment 10^{-7}, and seed 260815354.

A.4 Neural quadratic-form atlas

The solvable isotropic control integrates $\dot{\mathcal{M}} = Y\mathcal{M} + \mathcal{M}Y - 8c\mathcal{M}^2$ with $c = 1/8$ and $Y = 4m^{-1}\sum_\mu y_\mu A_\mu$, the symmetric label-weighted structure generator. The sign atlas represents distinct positive magnitudes by a geometric sequence from 1 to 0.25, distinct negative magnitudes by minus a geometric sequence from 0.35 to 1, and uses exact zeros. Each mode starts at 10^{-7} and its log-growth is fit at nine times in $[0, 0.2]$. The ignition experiment uses generator $\mathrm{diag}(1, .7, .7, .25, 0, -.4)$, width 32, 64 paired replicates, and $\epsilon = 10^{-2}, \ldots, 10^{-8}$. Gaussian, row-normalized, and row-Stiefel initializations are constructed from the same Gaussian draw in each replicate and then multiplied by $\sqrt{\epsilon}$. The half-plateau hitting time is solved to absolute tolerance 2×10^{-10}.

A.5 Two-dimensional settling-time atlas

The multiscale slice uses source model 1051 from `data/spectral_self_attention/continuous_ode_f2_chaos_antilock_matched.json`, with three planar tokens. Its score, value, and hidden-feature parameters are fixed; a scalar $\gamma = 1.4348$ multiplies the MLP bias, linear term, and quadratic output. The first two initial angles vary and the third is fixed at 1.4202138116039. Four 64×64 grids are integrated by RK4 with step 0.02 to time 220. The first covers $[-\pi, \pi)^2$. Each subsequent window is eight times narrower and is centered on the interior cell with the largest four-corner range of settling times in the preceding grid; the rule is deterministic. The zooms are adaptively selected and therefore do not form independent samples; their censored fractions cannot be read as population frequencies.

Settling time is one integration step after the last sample at which $\max_i |\dot{\theta}_i| \geq 10^{-5}$, and a trajectory is retained only if the final 20 time units remain below this threshold. Values that fail this test are right-censored. Terminal angle triples are clustered in wrapped ℓ_∞ distance at tolerance 10^{-3} as an endpoint audit; these finite-horizon clusters are not interpreted as attractor identities. A second audit retains a quiet endpoint only when it is closer to the numerically resolved stable synchronized root than to the companion saddle. The uncertainty calculation uses coordinate separations of 1, 2, 4, 8, and 16 pixels and thresholds $\delta \in \{1, 2, 5, 10\}$; its regressions condition on both endpoints being resolved. Each window also has a 32×32 spatial control at step 0.02 and

a matched step-0.01 control. The base and final windows are extended to time 300. A 24×24 base-grid scan over seven MLP multipliers separates the spatial test from the nearby parameter bifurcation. All coefficients, commands, censored values, terminal states, and pair counts are retained in the output directory. Counting the repeated numerical controls separately, the protocol integrates 30,656 initial conditions. The postprocessing also computes full three-angle finite-time tangent spectra for a typical trajectory, the longest resolved trajectory, and one trajectory below the fold, with step and perturbation controls. Pixel-strip geometry is measured before and after the root-distance audit. Analytic hyperbolic and critical-fold nulls use the measured unstable eigenvalue and quadratic coefficient; their fixed-window grids range from 64 to 4,096 points per coordinate, evaluated through the corresponding one-dimensional formulas rather than ODE integration.

B Status of the principal statements

Statement	Status	Scope or verification
Arbitrary attention-only equilibrium equation	Exact theorem	Any finite n, d, score S, value V, and $\beta \geq 0$; theorem 3.1.
Cluster multiplicity reduction	Exact identity	Covers every configuration with finitely many distinct token positions; (10).
Tied spectral–Gram system	Exact theorem	$S = V = V^{\mathsf{T}}$, including all signs, zero eigenvalues, and multiplicities; theorem 3.2.
Tied Lyapunov identity and exclusion of recurrence	Exact identity and consequence	Continuous attention-only flow under $S = V = V^{\mathsf{T}}$; (18).
No zero-output polygon for $V \succ 0$	Exact proposition	Tied exponential attention with $\beta > 0$; fails at $\beta = 0$.
Finite serial fixed-point and period-p equations	Exact equations	Defined whenever both normalization inputs are nonzero; (29) and (32).
Continuous-depth projected-sum equation	Controlled expansion	Fixed parameters on a compact neighborhood away from zero normalization inputs; local error $O(h^2)$.
Planar quadratic-MLP capacity	Exact proposition	Every angular trigonometric field of degree at most three; potential fields have zero constant mode.
Active-rotator and rotating-consensus orbits	Exact propositions	Explicit uniform- score and reducible two-head constructions; propositions 7.1 and 9.1.
Internal attention-only attracting cycle	Numerical evidence	Long RK4 integration, seven step ratios, full tangent spectrum, and 128 perturbations; no interval-arithmetic existence proof.
Two chaotic and one hyperchaotic attractor candidates	Numerical evidence	Complete finite-time spectra, divergence controls, step refinement, and basin replays; no proof of an invariant chaotic measure.
Finite-layer period-three and period-four census	Numerical observation	Frequencies are conditional on the stated random ensembles and tolerances.
Many-token transport and W_1 stability	Exact under regularity assumptions	Positive C^1 kernel and Lipschitz MLP on the compact sphere; (55).
NQF mode ignition and scaling tail	External theorem plus derived specialization	Requires the factorized NQF hypotheses, commuting/solvable reductions for ignition, and explicit power-law tails for the scaling law.
Synchronized saddle–node in the settling atlas	High-precision numerical calculation	The scalar reduction is exact; the critical parameter and normal-form coefficients are evaluated from stored decimal coefficients and are not interval-certified.

Statement	Status	Scope or verification
Near-saddle adiabatic condition	Local scaling argument	Smooth deterministic parameter paths in the scalar normal form; it is not a samplewise statement for Brownian OU paths; (72).
Smooth hyperbolic and fold settling-time nulls	Exact local calculations plus finite-grid evaluation	A hyperbolic saddle has asymptotic exponent one; the critical fold has exponent one half despite a smooth singular set; equations (47) and (50).
Fractal settling-time boundary in the atlas model	Not supported at the sampled scales	Four co-moving zooms, step/grid/horizon controls, root-distance endpoint audit, strip geometry, and finite-time tangent spectra. The general existence question outside this parameter slice remains open.

C Reproducibility map

The main experiments are available in the following files:

- finite-step inventory: `experiments/spectral_self_attention/large_scale_cycle_census.py`;
- fixed-point solving: `experiments/spectral_self_attention/equilibrium_root_census.py`;
- cycle validation: `experiments/spectral_self_attention/periodic_orbit_audit.py`;
- continuation down to $h/4096$: `experiments/spectral_self_attention/small_step_continuation.py` and `experiments/spectral_self_attention/extend_small_step_continuation.py`;
- continuous integration, robustness, and spectra: `experiments/spectral_self_attention/continuous_ode_audit.py`, `experiments/spectral_self_attention/continuous_ode_robustness.py`, and `experiments/spectral_self_attention/continuous_lyapunov_spectrum.py`;
- many-token and kernel limits: `experiments/spectral_self_attention/mean_field_extensions.py`;
- multihead and high dimension: `experiments/spectral_self_attention/multihead_phase.py`;
- first Muon step: `experiments/spectral_self_attention/one_step_muon.py`;
- slow OU weights: `experiments/spectral_self_attention/slow_ou_tokens.py`;
- NQF sign/multiplicity, ignition, structure-spectrum, tail, and BBP controls: `experiments/spectral_self_attention/nqf_training_spectrum_atlas.py`;
- two-dimensional settling-time and saddle-node audit: `experiments/spectral_self_attention/fractal_settling_atlas.py`;
- English figures for this manuscript: `experiments/spectral_self_attention/global_report_figures_en.py`.

The numerical aggregates used in the text are `data/spectral_self_attention/large_taxonomy_summary.json`, `data/spectral_self_attention/small_step_final_results.json`, `data/spectral_self_attention/mean_field_extensions/summary.json`, `data/spectral_self_attention/one_step_muon/summary.json`, `data/spectral_self_attention/multihead/frozen_probe_basins.json`, and `data/spectral_self_attention/slow_ou_tokens.json`. The settling-time atlas is stored under `data/spectral_self_attention/fractal_settling_atlas/`. The NQF atlas is stored under `data/spectral_self_attention/nqf_training_spectrum_atlas/`; its `data/spectral_self_attention/nqf_training_spectrum_atlas/summary.json` records the command configuration, errors, and twelve strict checks, while the CSV files retain every replicate.

At the stated version, rerunning the atlas with the configuration stored in `data/spectral_self_attention/nqf_training_spectrum_atlas/metadata.json` passes all twelve numerical checks. The focused regression suite comprising `tests/test_large_scale_cycle_census.py`, `tests/test_mean_field_extensions.py`, `tests/test_mlp_equilibrium_taxonomy.py`, `tests/test_multihead_phase.py`, the two `tests/test_nqf_*.py` files, `tests/test_one_step_muon.py`, and `tests/test_spectral_self_attention.py` passes 58 tests when invoked from the repository root with `PYTHONPATH=.`.

D Tangent stability criteria

For a fixed point, negative spectral abscissa of the tangent Jacobian implies local asymptotic stability, and a positive-real-part eigenvalue implies instability. A zero real part is inconclusive. A hyperbolic attracting periodic orbit has one zero Lyapunov exponent associated with phase and negative transverse exponents. A chaotic flow has a positive exponent and a zero flow exponent; hyperchaos has at least two positive exponents. Nonnormal tangent dynamics may produce finite-time amplification even when all asymptotic rates are negative. The numerical classification therefore combines tangent spectra, long trajectories, mean divergence, and repeated perturbations.

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